ESP32-S31

esp-dev-kits Documentation

Release master
Espressif Systems
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This document provides detailed user guides and examples for ESP32-S31 series development boards.

Note: For the full list of Espressif development boards, please go to ESP DevKits.

Chapter 1

ESP32-S31-Function-CoreBoard-1

The ESP32-S31-Function-CoreBoard-1 is a development board for connected AIoT prototyping based on the ESP32-S31-WROOM-3 module. It supports Wi-Fi 6, IEEE 802.15.4, Bluetooth 5.4 (LE + BR/EDR), and Gigabit Ethernet; provides an onboard microphone and speaker output; routes out all key GPIOs; and supports out-of-the-box AI voice interaction evaluation.

1.1 ESP32-S31-Function-CoreBoard-1

This user guide will help you get started with ESP32-S31-Function-CoreBoard-1 and provides detailed information about this development board.

The ESP32-S31-Function-CoreBoard-1 is a development board for connected AIoT prototyping based on the ESP32-S31-WROOM-3 module. It supports Wi-Fi 6, IEEE 802.15.4, Bluetooth 5.4 (LE + BR/EDR), and Gigabit Ethernet; provides an onboard microphone and speaker output; routes out all key GPIOs; and supports out-of-the-box AI voice interaction evaluation.

Most of the I/O pins on the module are broken out to pin header **J2** for easy interfacing.

The document consists of the following major sections:

- *Getting Started*: Overview of ESP32-S31-Function-CoreBoard-1 and hardware/software setup instructions to get started.
- *Hardware Reference*: More detailed information about the ESP32-S31-Function-CoreBoard-1 ' s hardware.
- *Hardware Revision Details*: Information about revision history, known issues, and links to user guides for previous versions (if any).
- *Related Documents*: Links to related documentation.
- *Disclaimer and Copyright Notice*: Link to the disclaimer and copyright notice.

1.1.1 Getting Started

This section briefly introduces ESP32-S31-Function-CoreBoard-1 and explains how to perform the initial hardware setup and how to flash firmware onto the board.

Description of Components

The following table lists the key components on the front and back of the board in clockwise order.

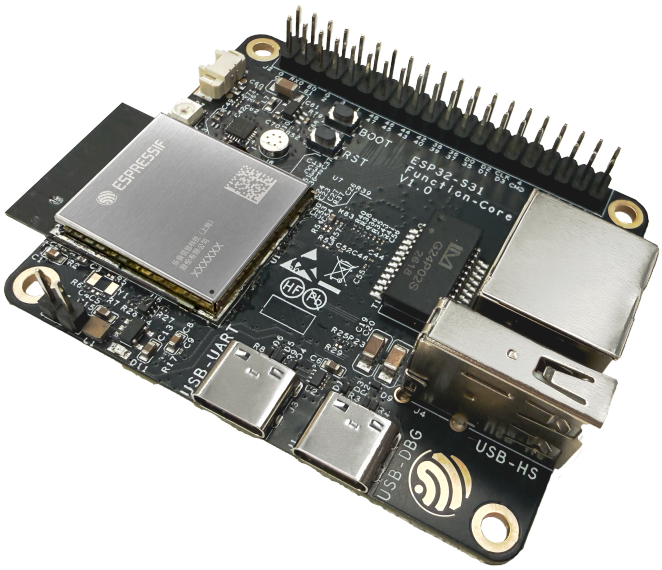

Fig. 1: ESP32-S31-Function-CoreBoard-1 (with ESP32-S31-WROOM-3 module on board)

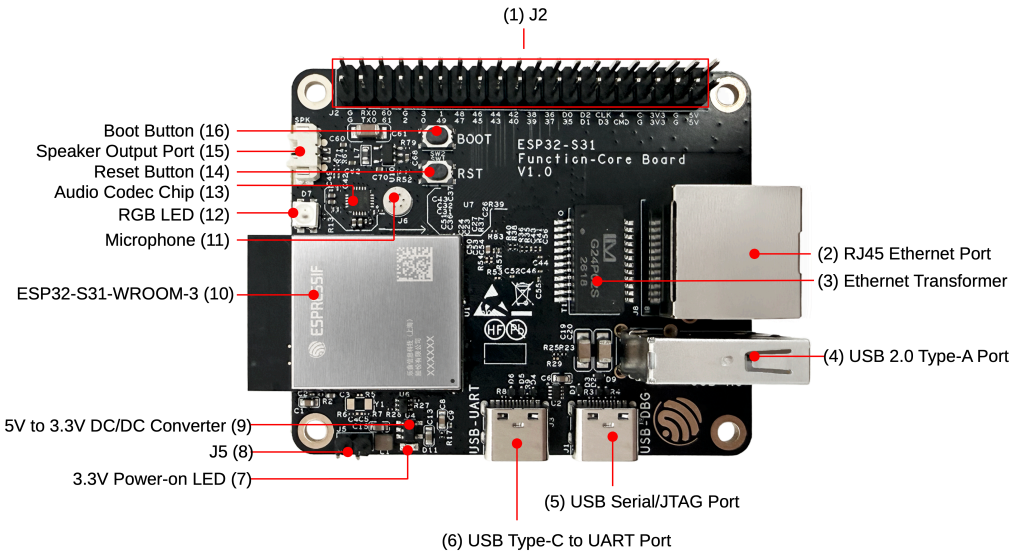

Fig. 2: ESP32-S31-Function-CoreBoard-1 - front (click to enlarge)

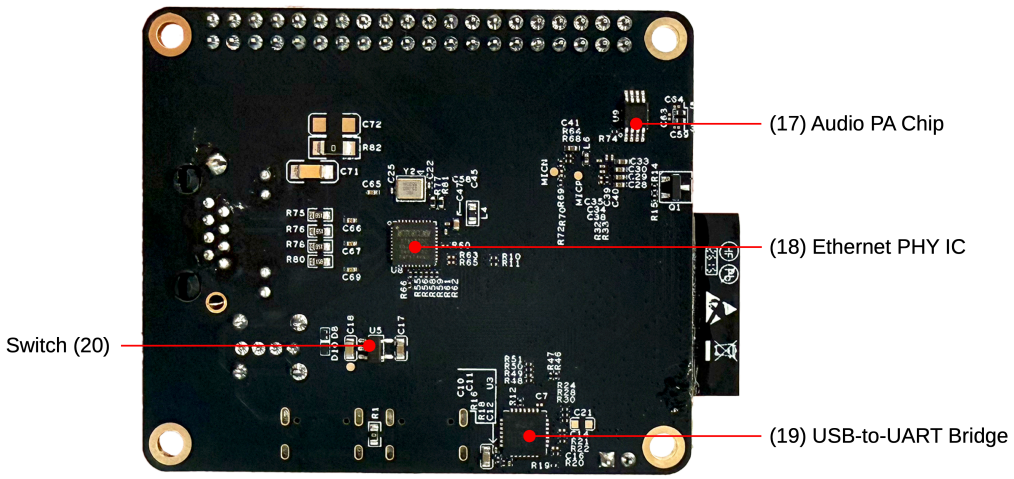

Fig. 3: ESP32-S31-Function-CoreBoard-1 - back (click to enlarge)

No.	Key Component	Description
1	J2	All available GPIO pins are broken out to header J2 for easy interfacing. For more details, see *Header Block*.
2	RJ45 Ethernet Port	An Ethernet port supporting 10/100/1000 Mbps auto-negotiation.
3	Ethernet Transformer	Transformer module for the RJ45 Ethernet port.
4	USB 2.0 Type-A Port	The USB 2.0 Type-A port is connected to the USB 2.0 OTG High-Speed interface of the ESP32-S31, compliant with the USB 2.0 specification. When communicating with other devices via this port, the ESP32-S31 acts as a USB host, providing up to 500 mA of current.
5	USB Serial/JTAG Port	USB Type-C port that supports USB 2.0 Full-Speed data rate. It can be used to supply power to the board, flash firmware to the ESP32-S31 chip, communicate with the chip via the USB protocol, and perform JTAG debugging.
6	USB Type-C to UART Port	Used for power supply to the board, flashing applications to the chip, as well as communication with the ESP32-S31 chip via the on-board USB-to-UART bridge.
7	3.3 V Power-on LED	Turns on when the board is connected to USB power.
8	J5	Used for current measurement. See details in section *Current Measurement*.
9	5 V to 3.3 V DC/DC Converter	Power regulator that converts a 5 V supply into a 3.3 V output.
10	ESP32-S31-WROOM-3	ESP32-S31-WROOM-3 integrates the ESP32-S31 chip and supports Bluetooth 5.4 (LE) and IEEE 802.15.4 (Zigbee/Thread/Matter), suitable for a wide range of low-power IoT applications.

No.	Key Component	Description
11	Microphone	Onboard microphone connected to the interface of the audio codec chip.
12	RGB LED	Addressable RGB LED, driven by GPIO60.
13	Audio Codec Chip	ES8311 is a low-power mono audio codec. It includes a single-channel ADC, a single-channel DAC, a low-noise pre-amplifier, a headphone driver, digital sound effects, analog mixing, and gain functions. It interfaces with the ESP32-S31 chip over I2S and I2C buses to provide hardware audio processing independent of the audio application.
14	Reset Button	Press this button to reset the ESP32-S31.
15	Speaker Output Port	This port is used to connect a speaker. The maximum output power can drive a 4 Ω, 3 W speaker. The pin spacing is 1.25 mm (0.08”).
16	Boot Button	Download button. Holding down **Boot** and then pressing **Reset** initiates Firmware Download mode for downloading firmware through the UART port or USB Serial/JTAG port.
17	Audio PA Chip	NS4150B is an EMI-compliant, 3 W mono Class D audio power amplifier that amplifies audio signals from the audio codec chip to drive speakers.
18	Ethernet PHY IC	Ethernet PHY chip connected to the ESP32-S31 RGMII interface and the RJ45 Ethernet port.
19	USB-to-UART Bridge	Single-chip USB-to-UART bridge that works with the **USB Type-C to UART Port** for board power, firmware flashing, and serial communication with the ESP32-S31 chip.
20	Switch	TPS2051C is a USB power switch that provides a 500 mA output current limit.

Start Application Development

Before powering up your ESP32-S31-Function-CoreBoard-1, please make sure that it is in good condition with no obvious sign of damage.

Required Hardware

- ESP32-S31-Function-CoreBoard-1
- USB 2.0 cable (USB-A to Type-C)
- Computer running Windows, Linux, or macOS

Note: Be sure to use a good quality USB cable. Some cables are for charging only and do not provide the needed data lines and will not work for programming this board.

Hardware Setup Connect the ESP32-S31-Function-CoreBoard-1 to your computer using a USB cable. The board can be powered through any of the USB Type-C ports.

Software Setup Please proceed to ESP-IDF Get Started, which will help you set up the development environment quickly and then flash an application onto your board.

Note: The board uses a USB port to communicate with the computer. Most operating systems (Windows, Linux, macOS) come with the required drivers pre-installed and the board is recognized automatically once plugged in. If the device cannot be recognized or a serial connection cannot be established, refer to Establish Serial Connection with ESP32-S31 for detailed driver installation steps.

Espressif provides board peripheral management components for many development boards to help you initialize and use key onboard peripherals such as LCD, audio codec, keys, and LEDs more easily and efficiently. Please visit the esp_board_manager component in ESP Component Registry for support coverage.

Other Development Framework Options In addition to ESP-IDF, this board supports the following frameworks and options for different user needs:

- Espressif Bluetooth LE ecosystem: Develop Bluetooth LE applications using solutions such as ESP-BLE-MESH and ESP-BLE-AUDIO to accelerate time-to-market and mass production.
- ESP-GMF: Espressif general multimedia framework with audio/video processing components for multimedia applications.
  - Wi-Fi audio/video examples: Sample Wi-Fi audio and video applications to help integrate multimedia features into your project.
  - Bluetooth audio: Unified Bluetooth audio APIs supporting Bluetooth Classic and LE Audio.
- ESP-Matter: Build Matter and Thread devices suited to low-power and battery-powered scenarios.

Contents and Packaging

Ordering Information The development board is available in the variant listed in the table below.

Part Number	On-board Module	Flash	PSRAM	Antenna
ESP32-S31-Function-CoreBoard-1	ESP32-S31-WROOM-3	16 MB	16 MB	PCB on-board antenna

For more information about available variants, visit the ESP Product Selector and select the **Development Board** tab. If you cannot find the variant you need in the table or Product Selector, or if you have special customization requirements, contact Espressif Sales.

Retail Orders If you order a few samples, each ESP32-S31-Function-CoreBoard-1 comes in an individual package in either an antistatic bag or any other packaging depending on your retailer.

For retail orders, please go to Get Samples.

Wholesale Orders If you order in bulk, the boards come in large cardboard boxes.

For wholesale orders, please go to Contact Us.

1.1.2 Hardware Reference

Block Diagram

The main components of ESP32-S31-Function-CoreBoard-1 and their interconnections are shown in the block diagram below.

Power Supply Options The board can be powered using one of the following options:

- Via the USB-to-UART port or the ESP32-S31 USB port (either port or both), default method (recommended)
- 5V and G (GND) pin headers

Current Measurement

The J5 headers on ESP32-S31-Function-CoreBoard-1 (see J5 in Figure *ESP32-S31-Function-CoreBoard-1 - front (click to enlarge)*) can be used for measuring the current drawn by the ESP32-S31-WROOM-3 module:

- Remove the jumper: Power supply between the module and peripherals on the board is cut off. To measure the module's current, connect an ammeter in series via the J5 headers.
- Apply the jumper (factory default): Restore the board's normal functionality.

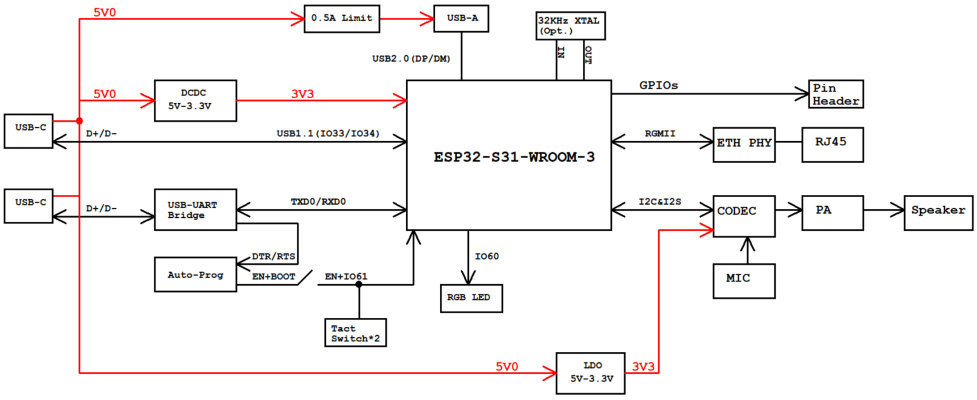

Fig. 4: ESP32-S31-Function-CoreBoard-1 block diagram (click to enlarge)

Header Block

The table below provides the **Name** and **Function** of pin header **J2**. The pin names are shown in Figure *ESP32-S31-Function-CoreBoard-1 - front (click to enlarge)*. The numbering is the same as in the ESP32-S31-Function-CoreBoard-1 schematic (PDF).

No.	Name	Type[Page 9, 1]	Function
1	G	G	Ground
2	G	G	Ground
3	TX0	I/O/T	U0TXD, GPIO58
4	RXD	I/O/T	U0RXD, GPIO59
5	61	I/O/T	BOOT, GPIO61
6	60	I/O/T	GPIO60[2]
7	2	I/O/T	GPIO2
8	G	G	Ground
9	0	I/O/T	GPIO0
10	3	I/O/T	GPIO3
11	49	I/O/T	GPIO49
12	1	I/O/T	GPIO1
13	47	I/O/T	GPIO47
14	48	I/O/T	GPIO48
15	45	I/O/T	GPIO45
16	46	I/O/T	GPIO46
17	43	I/O/T	GPIO43
18	44	I/O/T	GPIO44
19	40	I/O/T	GPIO40
20	42	I/O/T	GPIO42
21	39	I/O/T	GPIO39
22	38	I/O/T	GPIO38
23	37	I/O/T	GPIO37
24	36	I/O/T	GPIO36
25	35	I/O/T	GPIO35
26	D0	I/O/T	SDIO_DATA0, GPIO20
27	D1	I/O/T	SDIO_DATA1, GPIO21
28	D2	I/O/T	SDIO_DATA2, GPIO22
29	D3	I/O/T	SDIO_DATA3, GPIO23
30	CLK	I/O/T	SDIO_CLK, GPIO24

continues on next page

Table 1 – continued from previous page

No.	Name	Type[Page 9, 1]	Function
31	CMD	I/O/T	SDIO_CMD, GPIO25
32	4	I/O/T	GPIO4
33	G	G	Ground
34	G	G	Ground
35	3V3	P	3.3 V power supply
36	3V3	P	3.3 V power supply
37	G	G	Ground
38	G	G	Ground
39	5V	P	5 V power supply
40	5V	P	5 V power supply

J2

1.1.3 Hardware Revision Details

No previous versions available.

1.1.4 Related Documents

Please download the following documents from the HTML version of esp-dev-kits Documentation.

- ESP32-S31 Datasheet (PDF)
- ESP32-S31-WROOM-3 Datasheet (PDF)
- ESP32-S31-Function-CoreBoard-1 schematic (PDF)
- ESP32-S31-Function-CoreBoard-1 PCB layout (PDF)
- ESP32-S31-Function-CoreBoard-1 dimensions (PDF)
- ESP32-S31-Function-CoreBoard-1 dimensions source file (DXF) - You can view it with Autodesk Viewer online
- ESP32-S31-Function-CoreBoard-1 Reference Design (ZIP)

For further design documentation for the board, please contact us at sales@espressif.com.

1.1.5 Disclaimer and Copyright Notice

See *Disclaimer and Copyright Notice*.

[1] P: Power; I: Input; O: Output; T: High-impedance; G: Ground.
[2] Used to drive the addressable RGB LED (GPIO60).

Chapter 2

ESP32-S31-Korvo-1

The ESP32-S31-Korvo-1 is a multimedia development board based on the ESP32-S31-WROOM-3 module for smart audio and HMI applications. It features a dual-microphone array, speaker output, a microSD card slot, and an onboard DVP camera; supports speech recognition and voice wake-up; and comes with a 4.3-inch LCD subboard (ESP32-S3-LCD-EV-Board-SUB3). This document describes revision V1.1.

2.1 ESP32-S31-Korvo-1 V1.1

This user guide will help you get started with ESP32-S31-Korvo-1 V1.1 and provides detailed information about this development board.

The ESP32-S31-Korvo-1 V1.1 is a multimedia development board based on the ESP32-S31-WROOM-3 module for smart audio and HMI applications. It features a dual-microphone array, speaker output, a microSD card slot, and an onboard DVP camera; supports speech recognition and voice wake-up; and comes with a 4.3-inch LCD subboard (ESP32-S3-LCD-EV-Board-SUB3).

The ESP32-S31-Korvo-1 V1.1 main board can work with the LCD subboard ESP32-S3-LCD-EV-Board-SUB3. This document mainly introduces the ESP32-S31-Korvo-1 V1.1 main board. For more information about the LCD subboard, please refer to the ESP32-S3-LCD-EV-Board User Guide.

The document consists of the following major sections:

- *Getting Started*: Overview of ESP32-S31-Korvo-1 V1.1 and hardware/software setup instructions to get started.
- *Hardware Reference*: More detailed information about the ESP32-S31-Korvo-1 V1.1 hardware.
- *Hardware Revision Details*: Information about revision history, known issues, and links to user guides for previous versions.
- *Related Documents*: Links to related documentation.
- *Disclaimer and Copyright Notice*: Link to the disclaimer and copyright notice.

2.1.1 Getting Started

This section briefly introduces ESP32-S31-Korvo-1 V1.1 and explains how to perform the initial hardware setup and how to flash firmware onto the board.

Description of Components

The following list describes the key components on the board in a clockwise direction.

Fig. 1: ESP32-S31-Korvo-1 V1.1 (with ESP32-S31-WROOM-3 module on board)

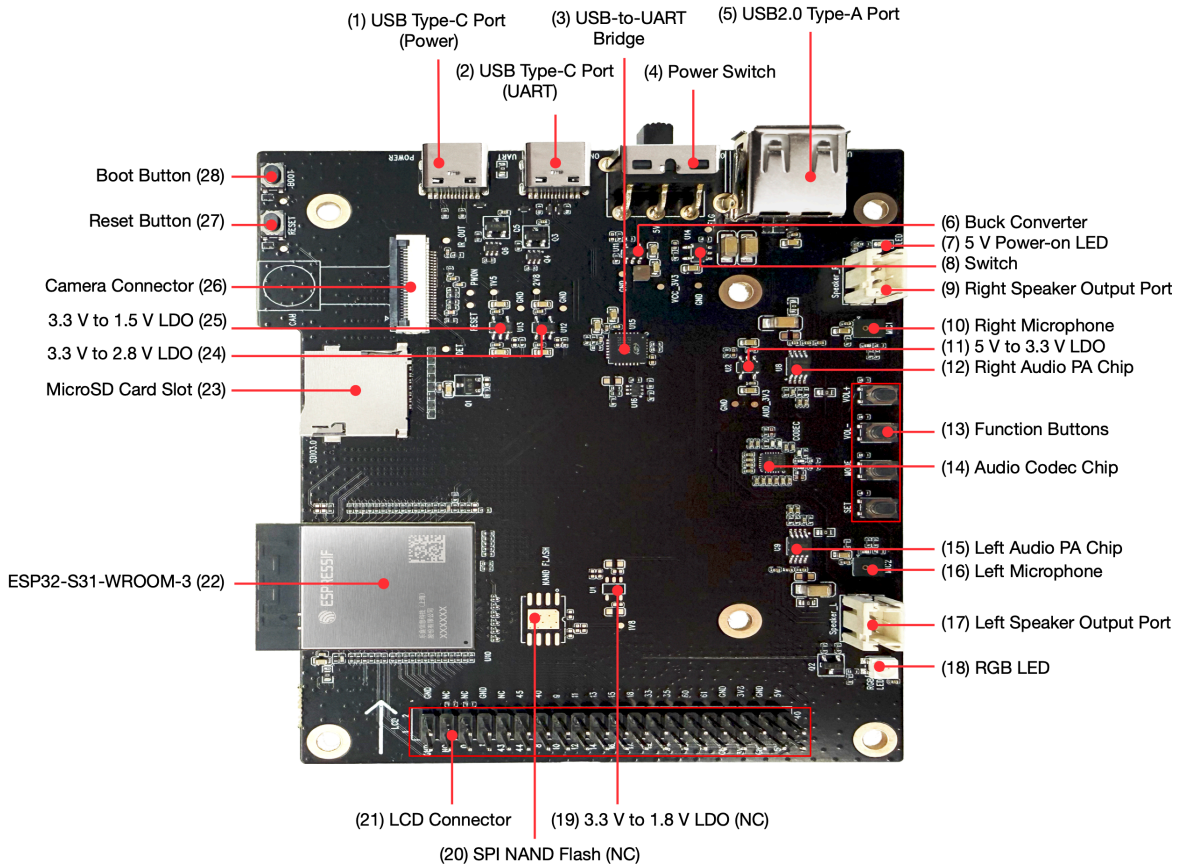

Fig. 2: ESP32-S31-Korvo-1 V1.1 (click to enlarge)

No.	Component	Description
1	USB Type-C Port (Power)	Power input only; no data communication.
2	USB Type-C Port (UART)	Can power the board, flash firmware to the chip, and communicate with the ESP32-S31 via the onboard USB-to-UART bridge.
3	USB-to-UART Bridge	Single-chip USB-to-UART bridge supporting up to 3 Mbps.
4	Power Switch	Slide toward ON to apply 5 V power; slide away from ON to disconnect 5 V power.
5	USB 2.0 Type-A Port	Connected to the ESP32-S31 USB 2.0 OTG High-Speed interface and supports the USB 2.0 standard. When used for USB communication, ESP32-S31 acts as a USB Host connected to downstream USB devices, providing up to 500 mA output current.
6	Buck Converter	Buck DC-DC converter for 3.3 V system power.
7	5 V Power-on LED	Lights when USB power is connected to the board.
8	Switch	TPS2051C USB power switch with 500 mA current limit.
9	Right Speaker Output Port	Right-channel speaker output; can drive a 4 Ω, 3 W speaker. The pin spacing is 2.00 mm (0.08”).
10	Right Microphone	Onboard right analog microphone routed to the audio codec.
11	5 V to 3.3 V LDO	Converts 5 V to 3.3 V for the audio circuitry.
12	Right Audio PA Chip	NS4150B low-EMI 3 W mono class-D amplifier driving the right speaker from the codec output.
13	Function Buttons	Four buttons: PLAY, SET, VOL-, and VOL+, connected to ESP32-S31-WROOM-3 for UI control and audio application testing via dedicated APIs.
14	Audio Codec Chip	ES8389 low-power stereo codec with dual ADC/DAC, low-noise preamp, headphone driver, digital effects, analog mixing, and gain control. Connected to the ESP32-S31 over I2S and I2C for hardware audio processing independent of application software.
15	Left Audio PA Chip	NS4150B low-EMI 3 W mono class-D amplifier driving the left speaker from the codec output.
16	Left Microphone	Onboard left analog microphone routed to the audio codec.
17	Left Speaker Output Port	Left-channel speaker output; can drive a 4 Ω, 3 W speaker. The pin spacing is 2.00 mm (0.08”).
18	RGB LED	Addressable RGB LED driven by GPIO37.

No.	Component	Description
19	3.3 V to 1.8 V LDO (NC)	Converts 3.3 V to 1.8 V for 1.8 V SPI NAND flash. Not populated by default (NC).
20	SPI NAND Flash (NC)	Quad SPI NAND flash sharing ESP32-S31-WROOM-3 signals with the microSD interface. Not populated by default (NC).
21	LCD Connector	Connector for the LCD subboard.
22	ESP32-S31-WROOM-3	ESP32-S31-WROOM-3 is a general-purpose module supporting 2.4 GHz Wi-Fi 6, Bluetooth 5.4, Bluetooth Classic, and IEEE 802.15.4 (Zigbee 3.0 and Thread 1.4). The module integrates ESP32-S31, 16 MB SPI flash, and 16 MB PSRAM, and uses an onboard PCB antenna.
23	microSD Card Slot	Supports 4-bit microSD for audio storage and playback. SDIO 3.0 capable.
24	3.3 V to 2.8 V LDO	Converts 3.3 V to 2.8 V for the camera module.
25	3.3 V to 1.5 V LDO	Converts 3.3 V to 1.5 V for the camera module.
26	Camera Connector	Connector for the onboard camera module for image transfer.
27	Reset Button	System reset.
28	Boot Button	Firmware download: hold Boot, press and release Reset to enter download mode; flash over the serial port.

Start Application Development

Before powering up the board, please make sure that it is in good condition with no obvious sign of damage.

Required Hardware

- ESP32-S31-Korvo-1 V1.1
- LCD subboard: ESP32-S3-LCD-EV-Board-SUB3
- One or two speakers
- Two USB 2.0 cables (Standard-A to Type-C)
- Computer running Windows, Linux, or macOS

Note: Be sure to use a good-quality USB cable. Some cables are for charging only and do not provide the needed data lines nor work for programming the board.

Optional Hardware

- microSD card

Hardware Setup

1. Connect the LCD subboard to the **LCD Connector**.
2. Connect one or two speakers to the **speaker output** port(s).
3. Plug in two USB cables from the PC to both USB ports on the board.
4. Turn on the **power switch**.
5. The red power LED should light.

Software Setup Please proceed to ESP-IDF Get Started, which will help you set up the development environment quickly and then flash an application onto your board.

To validate the board peripherals, you can also flash the factory demo .

Note: The board uses USB ports to communicate with the computer. Most operating systems (Windows, Linux, macOS) come with the required drivers pre-installed and the board is recognized automatically once plugged in. If the device cannot be recognized or a serial connection cannot be established, refer to Establish Serial Connection with ESP32-S31 for detailed driver installation steps.

Espressif provides board peripheral management components for many development boards to help you initialize and use key onboard peripherals such as LCD, audio codec, keys, and LEDs more easily and efficiently. Please visit the esp_board_manager component on the ESP Component Registry for support coverage.

Other Development Framework Options In addition to ESP-IDF, this board supports the following frameworks and options for different user needs:

- Espressif Bluetooth LE ecosystem: Develop Bluetooth LE applications using solutions such as ESP-BLE-MESH and ESP-BLE-AUDIO to accelerate time-to-market and mass production.
- ESP-Brookesia: Human-machine interaction framework for AIoT devices for graphical UI and smart display applications.
- ESP-GMF: Espressif general multimedia framework with audio/video processing components for multimedia applications.
  - Bluetooth audio: Unified Bluetooth audio APIs supporting Bluetooth Classic and LE Audio.
- ESP Video Components: Camera, video streaming, and video processing components for image capture and video applications.
- ESP-Matter: Build Matter and Thread devices suited to low-power and battery-powered scenarios.

Contents and Packaging

The package contains:

- ESP32-S31-Korvo-1 V1.1 main board (with the camera module installed)
- LCD subboard: ESP32-S3-LCD-EV-Board-SUB3

Ordering Information The development board is available in the variant listed in the table below.

Part Number	On-board Module	Flash	PSRAM	Antenna
ESP32-S31-Korvo-1	ESP32-S31-WROOM-3	16 MB	16 MB	PCB on-board antenna

For more information about available variants, visit the ESP Product Selector and select the **Development Board** tab. If you cannot find the variant you need in the table or Product Selector, or if you have special customization requirements, contact Espressif Sales.

Retail Orders If you order a few samples, each board comes in an individual package in either an antistatic bag or any other packaging depending on your retailer.

For retail orders, please go to Get Samples.

Wholesale Orders If you order in bulk, the boards come in large cardboard boxes.

For wholesale orders, please go to Contact Us.

2.1.2 Hardware Reference

Block Diagram

The main components of ESP32-S31-Korvo-1 V1.1 and their interconnections are shown in the block diagram below.

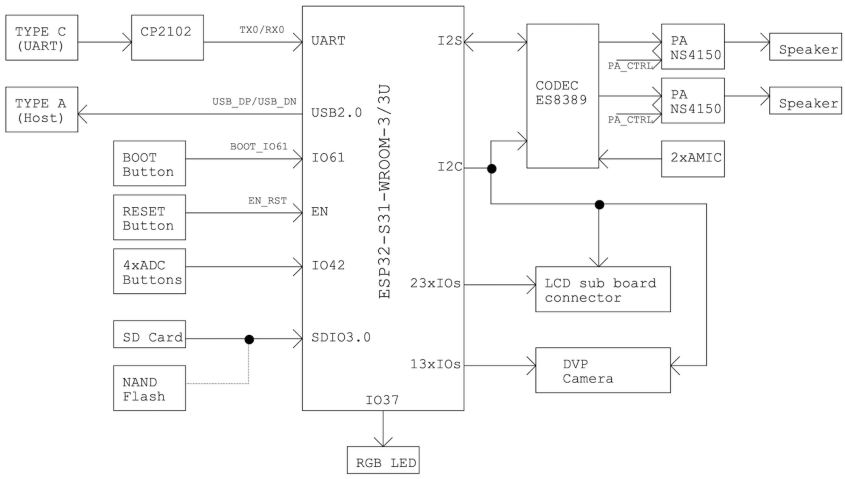

Fig. 3: ESP32-S31-Korvo-1 V1.1 electrical block diagram (click to enlarge)

Power Supply Description

USB Power Both USB Type-C ports can power the board: the Power port is power-only, while the UART port can supply power and carry data. When driving high-power speakers and using the USB Type-A port to power external devices at the same time, ensure the total input current to the board meets 3 A. Use a dedicated cable for USB power delivery, separate from the USB cable used for flashing applications.

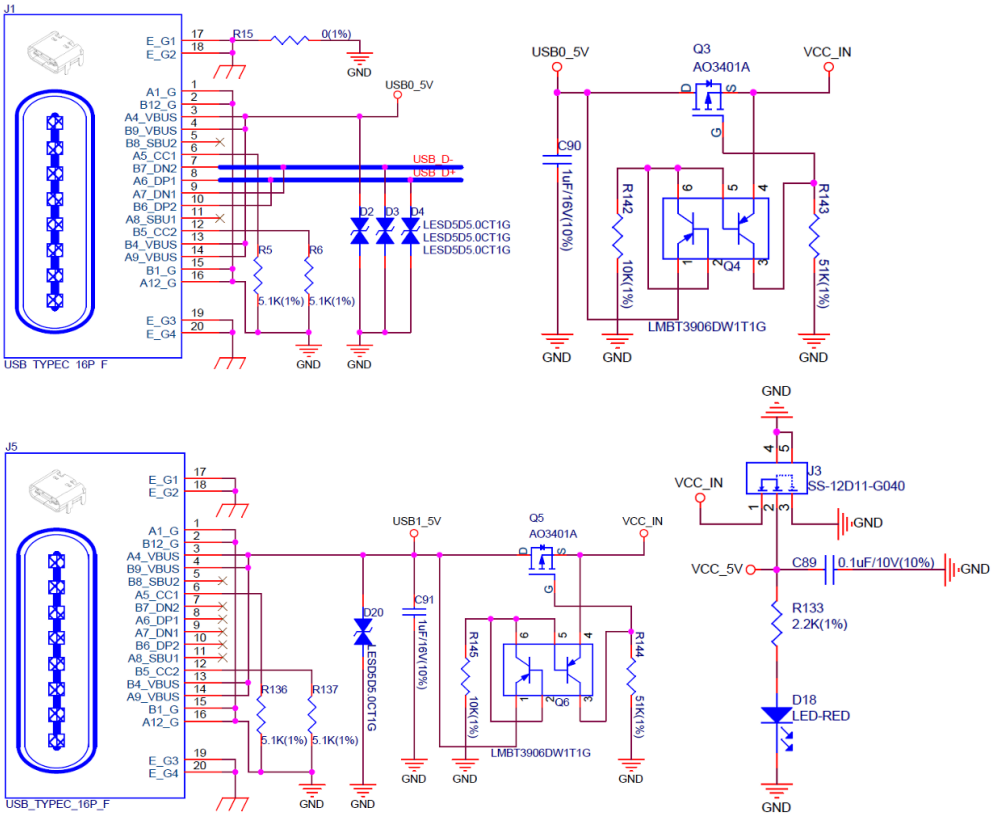

Fig. 4: ESP32-S31-Korvo-1 V1.1 - USB power supply (click to enlarge)

Separate Audio Power ESP32-S31-Korvo-1 V1.1 provides independent power supplies for audio components to reduce digital noise in audio signals and improve overall performance.

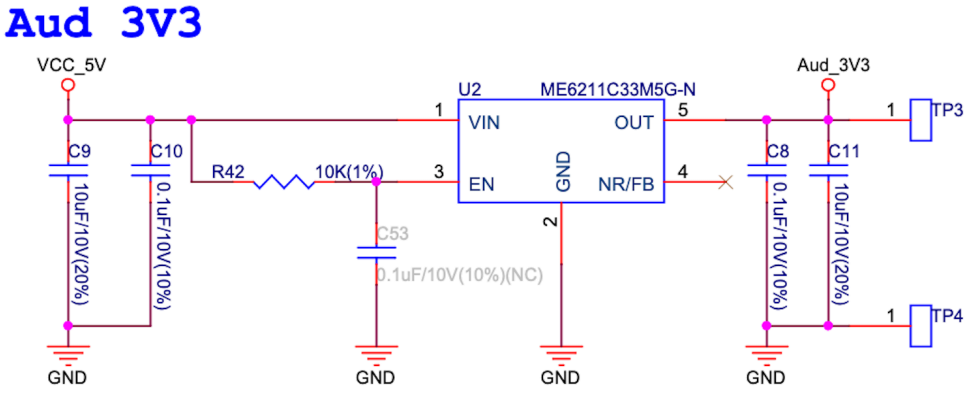

Fig. 5: ESP32-S31-Korvo-1 V1.1 - audio power supply (click to enlarge)

ES8389 Audio Codec Notes

ESP32-S31-Korvo-1 V1.1 uses the ES8389 audio codec. When transferring audio data with the SoC, note the following:

- Both DAC and ADC signals must be sent to the SoC in **TDM (Time Division Multiplexing)** format.
- **Pin 4 (MCLK)** of the ES8389 can be muxed to **TDMIN** via register configuration.

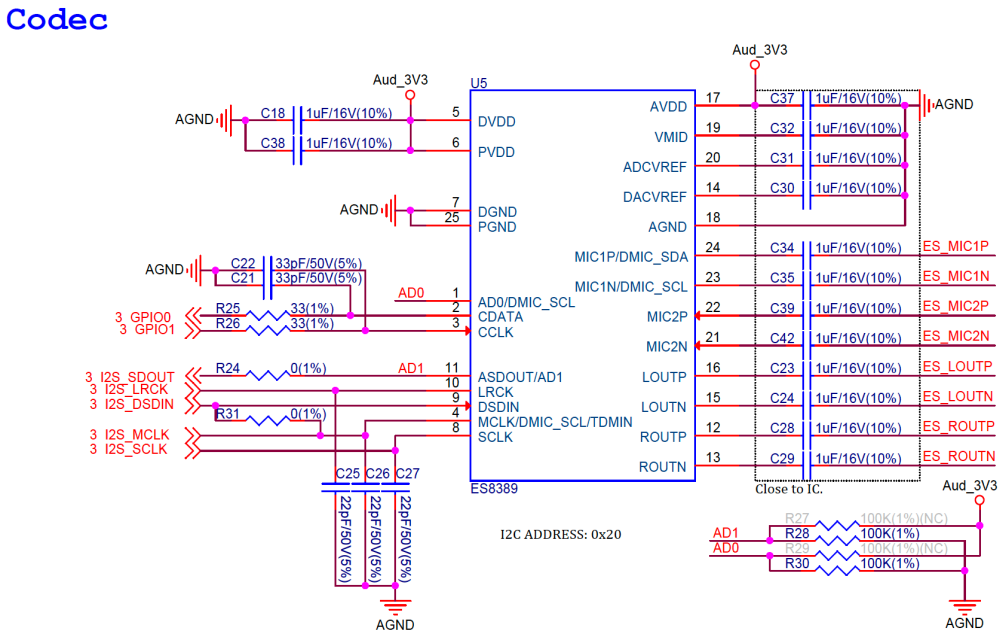

Fig. 6: ESP32-S31-Korvo-1 V1.1 - ES8389 audio codec schematic (click to enlarge)

microSD Card and SPI NAND Flash

The microSD card and SPI NAND flash functions share GPIO20–GPIO25 on the ESP32-S31-WROOM-3 module. The board uses the microSD card function by default. To switch to the SPI NAND flash function, perform hardware rework: remove R7, R65, R66, R67, R68, and R69, and populate R22, R23, R1, R2, R3, R4, C6, R20, and U4. Note that ESP32-S31 supports both 1.8 V and 3.3 V NAND flash. For **1.8 V** NAND flash, you must also populate R134, C66, C80, R100, U1, C82, and C67; for **3.3 V** NAND flash, populate R135.

Pin Assignment Table

The table below lists the ESP32-S31-WROOM-3 module signals routed to onboard peripherals, grouped by peripheral.

Table 1: ESP32-S31-WROOM-3 pin assignment

Peripheral	Signal	GPIO	Description
Audio I2S	I2S_MCLK	GPIO2	Master clock
	I2S_SCLK	GPIO3	Serial clock
	I2S_LRCLK	GPIO4	Frame clock (left/right channel select)
	I2S_SDOUT	GPIO5	Audio output data line (DAC playback)
	I2S_DSIN	GPIO6	Audio input data line (ADC recording)
Audio I2C	I2C_SDA	GPIO0	I2C data line for ES8389
	I2C_SCL	GPIO1	I2C clock line for ES8389
System	PA_CTRL	GPIO7	Enable control for left/right audio PA chips

continues on next page

Table 1 – continued from previous page

Peripheral	Signal	GPIO	Description
	WS2812_CTRL	GPIO37	Addressable RGB LED control
	ADC BUTTON	GPIO42	Function buttons (PLAY/SET/VOL-/VOL+) via resistor divider ADC
UART0	U0TXD	GPIO58	UART0 transmit (USB-to-UART bridge)
	U0RXD	GPIO59	UART0 receive (USB-to-UART bridge)
microSD / SPI NAND	SDIO_DATA0	GPIO20	Default microSD; SPI2_CLK when switched to SPI NAND
	SDIO_DATA1	GPIO21	Default microSD; SPI2_D when switched to SPI NAND
	SDIO_DATA2	GPIO22	Default microSD; SPI2_Q when switched to SPI NAND
	SDIO_DATA3	GPIO23	Default microSD; SPI2_CS when switched to SPI NAND
	SDIO_CLK	GPIO24	Default microSD; SPI2_HOLD when switched to SPI NAND
	SDIO_CMD	GPIO25	Default microSD; SPI2_WP when switched to SPI NAND
	SD_CTRL	GPIO39	microSD power/switch control; also Boot Mode 1
USB 2.0	USB2.0_DP	USB_DP	USB 2.0 D+
	USB2.0_DM	USB_DM	USB 2.0 D-
RGB LCD	DB0 (B3)	GPIO8	LCD data bit 0
	DB1 (B4)	GPIO9	LCD data bit 1
	DB2 (B5)	GPIO10	LCD data bit 2
	DB3 (B6)	GPIO11	LCD data bit 3
	DB4 (B7)	GPIO12	LCD data bit 4
	DB5 (G2)	GPIO13	LCD data bit 5
	DB6 (G3)	GPIO14	LCD data bit 6
	DB7 (G4)	GPIO15	LCD data bit 7
	DB8 (G5)	GPIO16	LCD data bit 8
	DB9 (G6)	GPIO17	LCD data bit 9
	DB10 (G7)	GPIO18	LCD data bit 10
	DB11 (R3)	GPIO19	LCD data bit 11
	DB12 (R4)	GPIO33	LCD data bit 12
	DB13 (R5)	GPIO34	LCD data bit 13
	DB14 (R6)	GPIO35	LCD data bit 14
	DB15 (R7)	GPIO36	LCD data bit 15
	LCD_CS	GPIO38	LCD chip select; also Boot Mode 0 and GM_FK
	LCD_PCLK	GPIO40	LCD pixel clock; also Boot Mode 2
	LCD_H_EN	GPIO43	LCD data enable
	LCD_H_SYNC	GPIO44	LCD horizontal sync
	LCD_V_SYNC	GPIO45	LCD vertical sync
	LCD_MOSI	GPIO60	LCD SPI MOSI; also Boot Mode 3
	LCD_SCK	GPIO61	LCD SPI clock; also Boot Mode 4
DVP camera	CAM_D0	GPIO46	Camera data bit 0
	CAM_D1	GPIO47	Camera data bit 1
	CAM_D2	GPIO48	Camera data bit 2
	CAM_D3	GPIO49	Camera data bit 3
	CAM_D4	GPIO50	Camera data bit 4
	CAM_D5	GPIO51	Camera data bit 5
	CAM_D6	GPIO52	Camera data bit 6
	CAM_D7	GPIO53	Camera data bit 7
	CAM_PCLK	GPIO54	Camera pixel clock
	CAM_XCLK	GPIO55	Camera master clock

continues on next page

Table 1 – continued from previous page

Peripheral	Signal	GPIO	Description
	CAM_V_SYNC	GPIO56	Camera vertical sync
	CAM_H_SYNC	GPIO57	Camera horizontal sync
	GM_FK	GPIO38	Camera GM_FK; shared with LCD_CS and Boot Mode 0

Hardware Setup Options

Automatic Download There are two ways to put the ESP development board into download mode:

- Manually press the Boot and RST buttons, then release RST first and Boot afterwards.
- Let software perform automatic download using the serial port DTR and RTS signals to control EN and GPIO61. For details, see the ESP32-S31-Korvo-1 V1.1 schematic (PDF).

2.1.3 Hardware Revision Details

- ESP32-S31-Korvo-1 V1.1:
  Matte black solder mask; larger PCB footprint; LCD subboard stacks on top of the main board when assembled. GPIO mapping is unchanged from V1.0.
- ESP32-S31-Korvo-1 V1.0:
  First revision with green solder mask; LCD subboard extended beyond the board edge so onboard functions remained exposed for debugging.

2.1.4 Related Documents

Please download the following documents from the HTML version of esp-dev-kits Documentation.

- ESP32-S31 Datasheet (PDF)
- ESP32-S31-WROOM-3 Datasheet (PDF)
- ESP32-S31-Korvo-1 V1.1 schematic (PDF)
- ESP32-S31-Korvo-1 V1.1 PCB layout (PDF)
- ESP32-S31-Korvo-1 V1.1 dimensions (PDF)
- ESP32-S31-Korvo-1 V1.1 dimensions source file (DXF) - You can view it with Autodesk Viewer online
- ESP32-S31-Korvo-1 V1.1 3D Printed Case Files (STL & STEP) - 3D printable case files are available here
- ESP32-S31-Korvo-1 Reference Design (ZIP)

For further design documentation for the board, please contact us at sales@espressif.com.

2.1.5 Disclaimer and Copyright Notice

See *Disclaimer and Copyright Notice*.

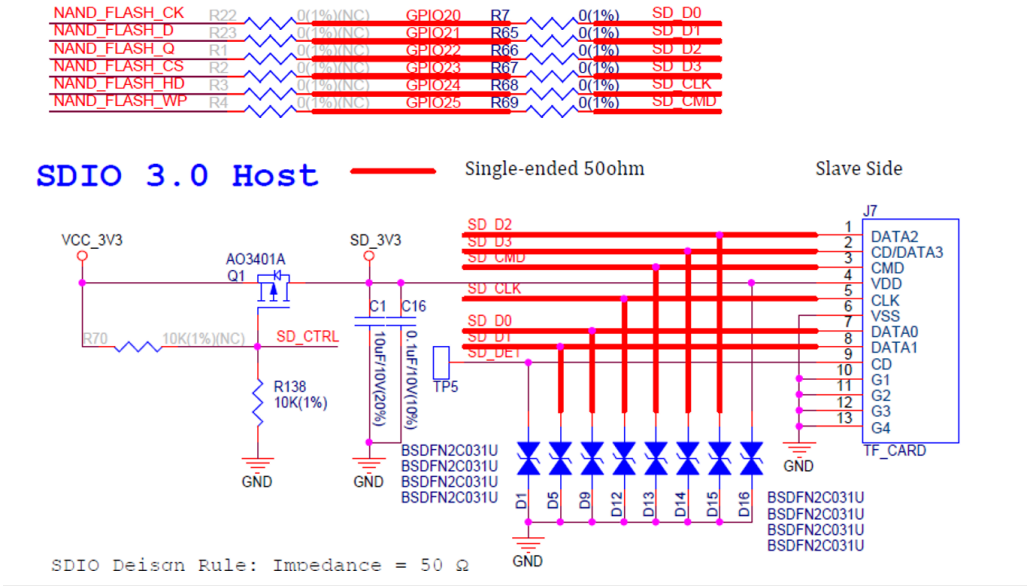

Fig. 7: ESP32-S31-Korvo-1 V1.1 - microSD card function (click to enlarge)

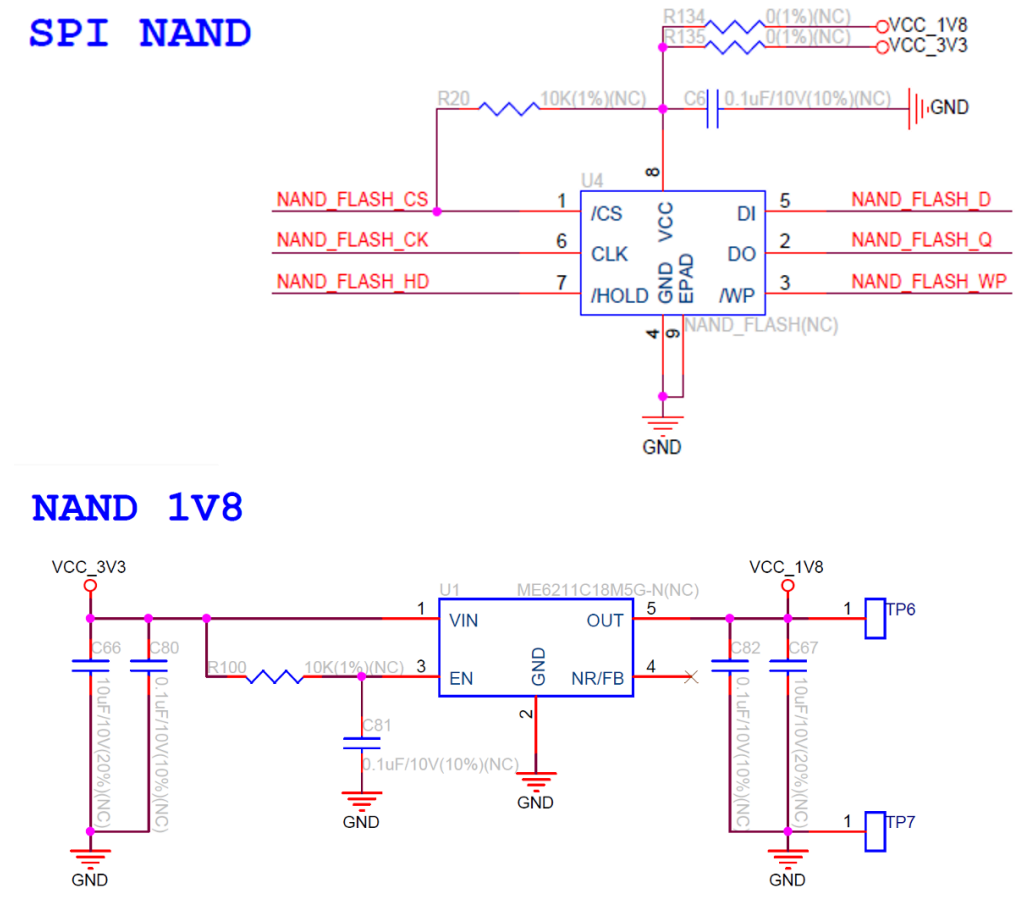

Fig. 8: ESP32-S31-Korvo-1 V1.1 - SPI NAND flash function (click to enlarge)

Chapter 3

ESP-Mosaico

ESP-Mosaico is an expandable smart interaction development kit based on ESP32-S31. It targets magnetic expansion, motion sensing, square touch display, and edge multimedia scenarios. The device integrates an ESP32-S31 SoC, a 480 × 480 QSPI square touch display, audio codec and amplifier, a 6-axis IMU, dual magnetometers, SPI NAND flash, and left/right 2 × 10P, 2.54 mm pitch module headers for camera and other expansions.

3.1 Reading Guide

Choose the documentation that matches your hardware and intended use:

- **CoreBoard hardware and development**: ESP-Mosaico has two CoreBoard versions, V1.2 and V1.0. Check the version on the CoreBoard silkscreen, then read the *User Guide (V1.2)* or the *User Guide (V1.0)* for components, interfaces, pin assignments, and development preparation.
- **Expansion modules**: When using the camera module or multi-function interaction module, read the *Camera Module User Guide* or *Multi-function Interaction Module User Guide* alongside the corresponding mainboard guide. Both module guides apply to CoreBoard V1.2 and V1.0. Module PCB versions and mainboard versions must be identified separately.
- **Factory application operation and firmware updates**: For button operation, expansion features, and factory application usage, refer to the external ESP-Mosaico Factory Application Guide. Use the Online Firmware Update page to flash firmware in the browser. That website focuses on application operation, while the guides below provide hardware information and development references.

3.2 Documentation

3.2.1 ESP-Mosaico V1.2

Older version: *ESP-Mosaico V1.0*

Note: This document covers CoreBoard **V1.2** hardware. Please check the silkscreen version on the mainboard to confirm your board revision. See *Hardware Revision Details* for differences from V1.0.

The camera module is supported in the **left slot only** (`H2`). Do not insert it into the right slot. For installation and usage instructions, see the *Camera Module User Guide*.

This guide helps you get started with ESP-Mosaico and provides detailed information about the board.

ESP-Mosaico is an expandable smart interaction development kit based on ESP32-S31. It targets magnetic expansion toys, motion sensing, square touch display HMI, and edge voice/multimedia applications. The device integrates a 480 × 480 QSPI square touch display, ES8311 audio codec, NS4150B amplifier, BMI270 6-axis IMU, dual BMM150 magnetometers, SPI NAND flash, and left/right 2 × 10P, 2.54 mm pitch module headers.

The main controller is Espressif ESP32-S31, supporting 2.4 GHz Wi-Fi 6 (IEEE 802.11b/g/n/ax), Bluetooth® 5.4 (LE + BR/EDR), and IEEE 802.15.4 (Thread / Zigbee). For storage, the board provides 16 MB in-package PSRAM, onboard 16 MB SPI NOR flash, and onboard 1 Gbit SPI NAND flash for graphics, multimedia, and asset caching. The display uses a CO5300-driven 480 × 480 QSPI square touch display with a CST9220-series touch controller.

For audio, ESP-Mosaico integrates ES8311 and NS4150B Class-D PA. The power system supports USB Type-C 5 V and a 3.7 V / 65 mAh Li-ion battery, with TP4057 charging, BQ27220 fuel gauge, and SAM8108 power-switch control. Both the Type-C port and the left/right module interfaces support power input and output. A USB 2.0 High-Speed OTG port supports power, debug, and communication, and the board also reserves UART expansion capability and left/right module interfaces.

Fig. 1: ESP-Mosaico Appearance (Click to enlarge)

The document consists of the following major sections:

- *Getting Started*: Overview of the board and hardware/software setup instructions.
- *Hardware Reference*: More detailed information about the board hardware.
- *Hardware Revision Details*: Hardware revision history and known issues (if any).
- *Related Documents*: Links to related documentation.
- *Disclaimer and Copyright Notice*: Link to the disclaimer and copyright notice.

Getting Started

This section briefly introduces ESP-Mosaico and describes how to flash firmware and prepare the board.

Description of Components ESP-Mosaico consists of a CoreBoard and a BaseBoard. The front views and their main components are shown first, followed by the back views and their main components.

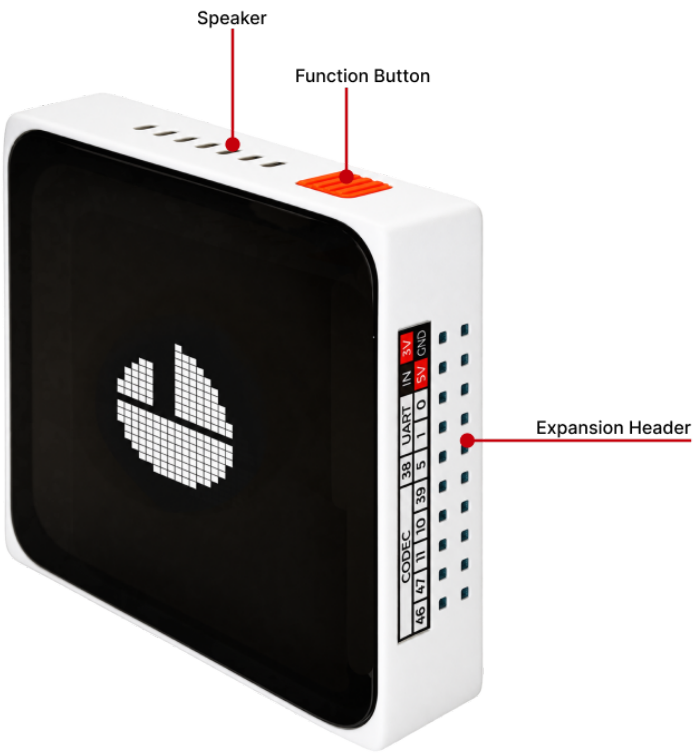

Fig. 2: ESP-Mosaico Front (Click to enlarge)

Front The following describes the main components on the front PCB in clockwise order.

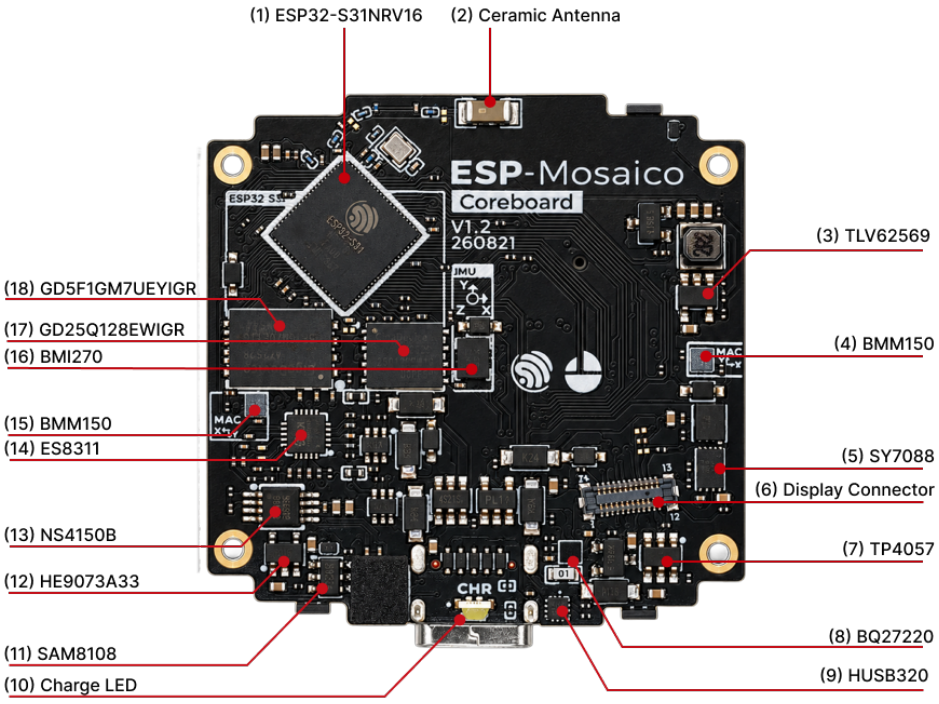

Fig. 3: ESP-Mosaico CoreBoard Front (Click to enlarge)

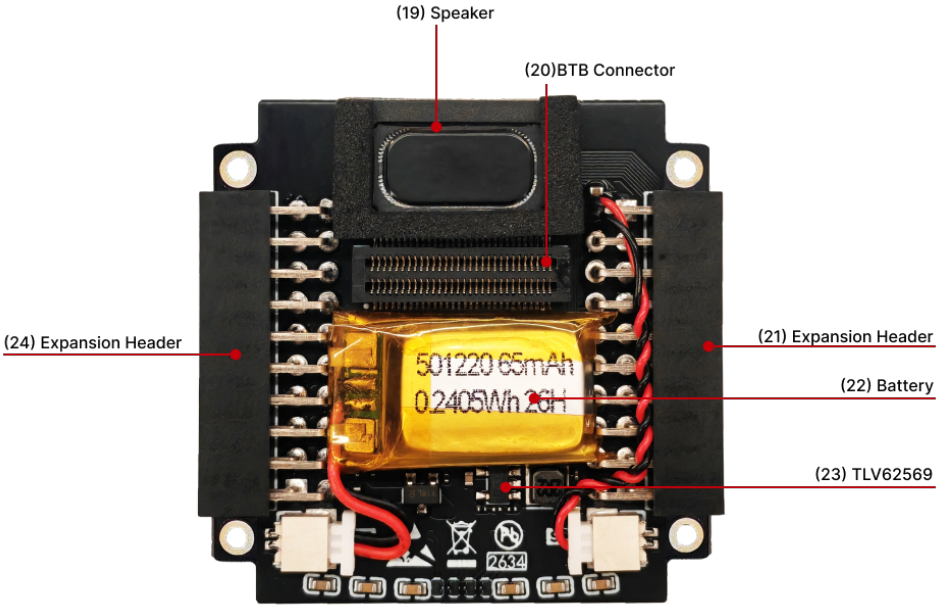

Fig. 4: ESP-Mosaico BaseBoard Front (Click to enlarge)

No.	Key Component	Description
	CoreBoard	Core board integrating the SoC, RF antenna, power management, display connector, audio codec, and sensors.
1	ESP32-S31NRV16	ESP32-S31 SoC, RISC-V 32-bit dual-core up to 320 MHz with 16 MB in-package PSRAM; supports 2.4 GHz Wi-Fi 6, Bluetooth 5.4 (LE + BR/EDR), and IEEE 802.15.4.
2	KH-3216-A35 Ceramic Antenna	2.4 GHz chip ceramic antenna for Wi-Fi and Bluetooth RF.
3	TLV62569	Buck (DCDC) converter that converts the input supply to the system 3.3 V rail for the display.
4	BMM150	3-axis magnetometer (BMM150 #2), I2C address 0x11.
5	SY7088	Boost (BOOST) converter for rails higher than the battery voltage.
6	Display Connector	Connects the 480 × 480, 2.16-inch QSPI touch display (CO5300 driver + CST9220 touch).
7	TP4057	Li-ion battery charger for the onboard battery.
8	BQ27220	Fuel gauge at I2C address 0x55 for voltage, current, state of charge (SOC), and related status.
9	HUSB320	USB Type-C interface controller that handles CC detection and controls the USB power direction.
10	Charge LED	Charge status indicator: orange while charging, white when fully charged.
11	SAM8108	Power on/off controller used with the POWER button.
12	HE9073A33	LDO providing a dedicated supply to the audio codec to reduce noise.
13	NS4150B	Low-EMI, filterless 3 W mono Class-D amplifier for the speaker.
14	ES8311	Low-power mono audio codec, I2C address 0x19, connected to the SoC over I2S.
15	BMM150	3-axis magnetometer (BMM150 #3), I2C address 0x12.
16	BMI270	6-axis IMU (accelerometer + gyroscope), I2C address 0x69.
17	GD25Q128EWIGR	16 MB (128 Mbit) SPI NOR flash for program and data storage.
18	GD5F1GM7UEYIGR	1 Gbit (128 MB) SPI NAND flash for large-capacity asset storage.
	BaseBoard	Base board integrating the speaker, battery, external 3.3 V buck, and left/right 2 × 10P module headers, mated to the CoreBoard through a board-to-board connector.
19	Speaker	Onboard speaker driven by the NS4150B amplifier.
20	BTB Connector	60-pin 0.5 mm board-to-board connector linking the CoreBoard and BaseBoard.
21	Expansion Header	2 × 10P dual-row, 2.54 mm pitch header on the right side of the figure, exposing GPIOs and power for expansion.
22	Battery	3.7 V / 65 mAh Li-ion battery for portable power.
23	TLV62569	Buck (DCDC) converter that generates the external 3.3 V rail, isolated from the internal MCU supply.
24	Expansion Header	The other header, on the left side of the figure, exposing the remaining GPIOs and power.

Back The following describes the main components on the back PCB in clockwise order.

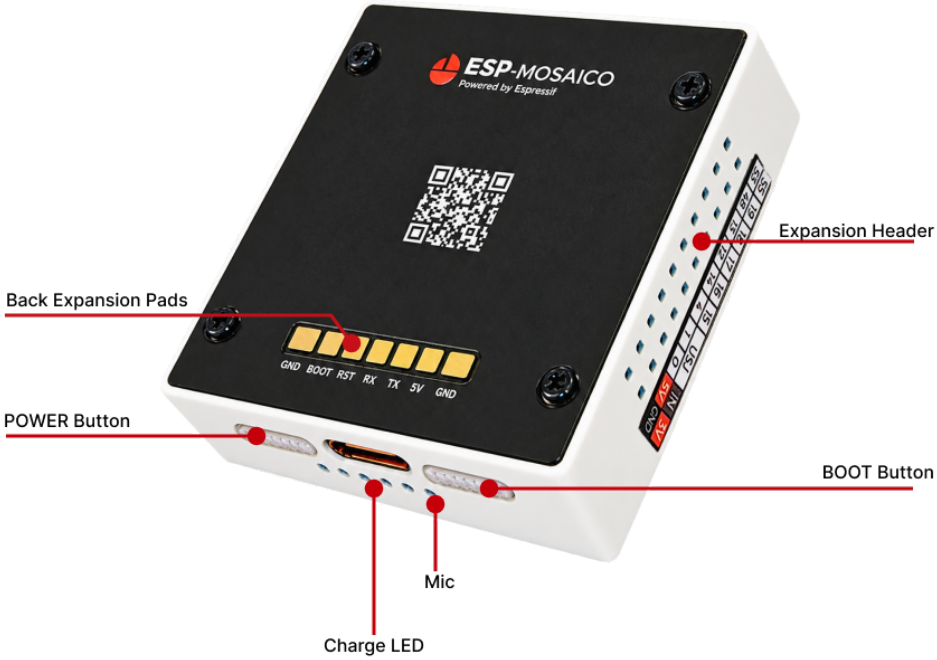

Fig. 5: ESP-Mosaico Back (Click to enlarge)

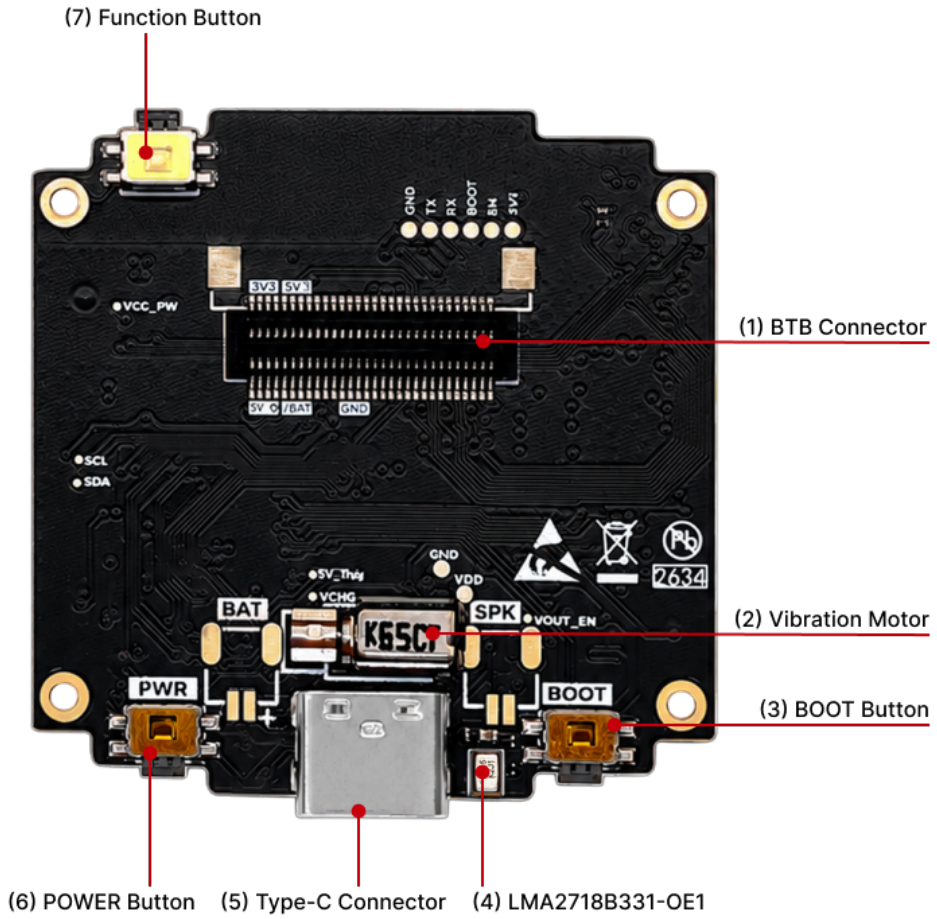

Fig. 6: ESP-Mosaico CoreBoard Back (Click to enlarge)

Fig. 7: ESP-Mosaico BaseBoard Back (Click to enlarge)

No.	Key Component	Description
	CoreBoard	Core board back side, exposing the buttons, Type-C connector, microphone, and vibration motor.
1	BTB Connector	Board-to-board connector mating with the BaseBoard (back side).
2	Vibration Motor	Vibration motor controlled by GPIO8 (active-high) for haptic feedback.
3	BOOT Button	Hold during power-up to enter firmware download mode.
4	LMA2718B331-OE1	Onboard microphone connected to the ES8311 for voice capture.
5	Type-C Connector	USB 2.0 High-Speed interface for power, debug/download, and USB communication, with Li-ion charging support.
6	POWER Button	Single press toggles the device power on/off.
7	Function Button	Application button (AI button) on GPIO7, active-low.
	BaseBoard	Base board back side, providing the back expansion pads.
8	Back Expansion Pads	One row of pads silkscreened GND, BOOT, RST, RX, TX, 5V, and GND. They can be used for debug, and the 5V pad can also supply the device.

Start Application Development Before powering up, make sure ESP-Mosaico is in good condition.

Required Hardware

- ESP-Mosaico
- USB cable (data-capable)
- Computer (Windows, Linux, or macOS)

Note: Be sure to use a proper USB data cable. Some cables are charge-only and cannot be used for data transfer or programming.

Hardware Notes

1. Flashing
   Connect ESP-Mosaico to the computer with a USB cable through the onboard Type-C port. This port is USB 2.0 High-Speed and supports both power delivery and USB data communication. The native ESP32-S31 chip only supports firmware download through this port in download mode, and does not provide auto-download or log output by itself. The factory firmware integrates a CDC virtual serial port and supports auto-download and serial log output. If flashing fails, try holding the BOOT button while the device is powered off, then power it on to enter download mode. In addition, the left and right module interfaces provide USB Serial/JTAG (left) and a TX/RX UART (right), both of which can be used for debugging and firmware download.
2. Power
   The Type-C port can power the device. The onboard battery is charged at a rated current of 65 mA. When a USB sink device is attached, the Type-C port automatically switches to source mode, with a maximum output of 5 V / 500 mA. If the load exceeds 1 A, the port shuts off automatically and recovers after a period of time. The left and right module interfaces also provide outward 3.3 V and 5 V rails (`3V` / `5V`). These rails are output only when GPIO60 is driven low. `VIN` is an input supply pin that can also charge the onboard battery.

Software Setup Follow the ESP-IDF Get Started guide to set up the development environment. The target chip for this board is `esp32s31`.

Note: The board communicates with the PC over USB. Most operating systems (Windows, Linux, macOS) already include the required drivers. If the device is not recognized or the serial connection fails, see Establish Serial Connection with ESP32-S31 for driver installation details.

BSP and Examples Espressif provides a board support package (BSP) for ESP-Mosaico to simplify initialization of the LCD, touch, audio, sensors, battery, NAND flash, module resources, and USB console.

- For the shipped application, buttons, expansion I/O, and getting started, see the ESP-Mosaico User Guide.
- Flash the latest firmware in the browser at Firmware Update.
- Recommended application framework: ESP-Claw.
- After the first successful flash, use the BSP USB CDC console and auto-download support. Typical commands:

```
idf.py set-target esp32s31
idf.py -p /dev/ttyACM0 flash monitor
```

Note:

- The first flash may require manually entering ROM download mode.
- The USB device briefly disconnects while the chip resets; IDF Monitor usually reconnects automatically.
- Use `monitor --no-reset` when attaching without restarting the application.
- Related options: `CONFIG_BSP_USB_CONSOLE`, `CONFIG_BSP_USB_CONSOLE_AUTO_INIT`, `CONFIG_BSP_USB_AUTO_DOWNLOAD`.

Contents and Packaging

Ordering Information The development board is available in the variant listed in the table below.

Part Number	On-board Module	Flash	PSRAM	Antenna
ESP-Mosaico	—	16 MB	16 MB	PCB on-board antenna

For more information about available variants, visit the ESP Product Selector and select the **Development Board** tab. If you cannot find the variant you need in the table or Product Selector, or if you have special customization requirements, contact Espressif Sales.

Retail Orders If you order a few samples, each board comes in an individual package in either antistatic bag or any packaging depending on your retailer.

For retail orders, please go to https://www.espressif.com/en/company/contact/buy-a-sample.

Wholesale Orders If you order in bulk, the boards come in large cardboard boxes.

For wholesale orders, please go to https://www.espressif.com/en/contact-us/sales-questions.

Hardware Reference

Block Diagram The main components and connections of ESP-Mosaico are shown below.

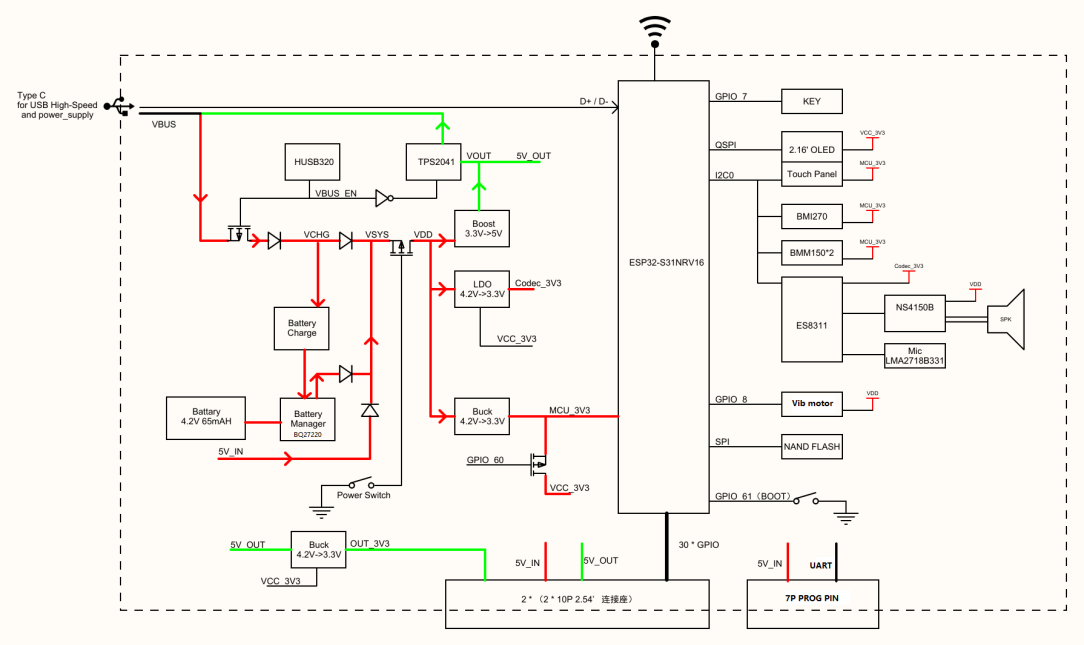

Fig. 8: ESP-Mosaico Block Diagram (Click to enlarge)

Power Supply Options You can power the board in the following ways:

1. Through the `Type-C (USB-C)` port
   Connect a USB Type-C cable to the Type-C port.
2. Through the `back expansion pads`
   Supply external power through the back expansion pads on the BaseBoard.
3. Through the `battery`
   The board supports a 3.7 V / 65 mAh Li-ion battery.

Regardless of the power source, use the POWER button to turn the device on or off. External power sources above can also charge the onboard battery.

Type-C Interface The Type-C port connects to ESP32-S31 USB 2.0 High-Speed OTG for power, debug, and application communication. Onboard USB Serial/JTAG signals are used for download and debugging. The BSP USB CDC console and auto-download feature are the recommended download and log path.

DCDC 3.3 V Circuit TLV62569 steps `VDD` down to `MCU_3V3`, which is then switched to `VCC_3V3` by the `VCC_PW` load switch.

Power-control GPIOs defined by the BSP:

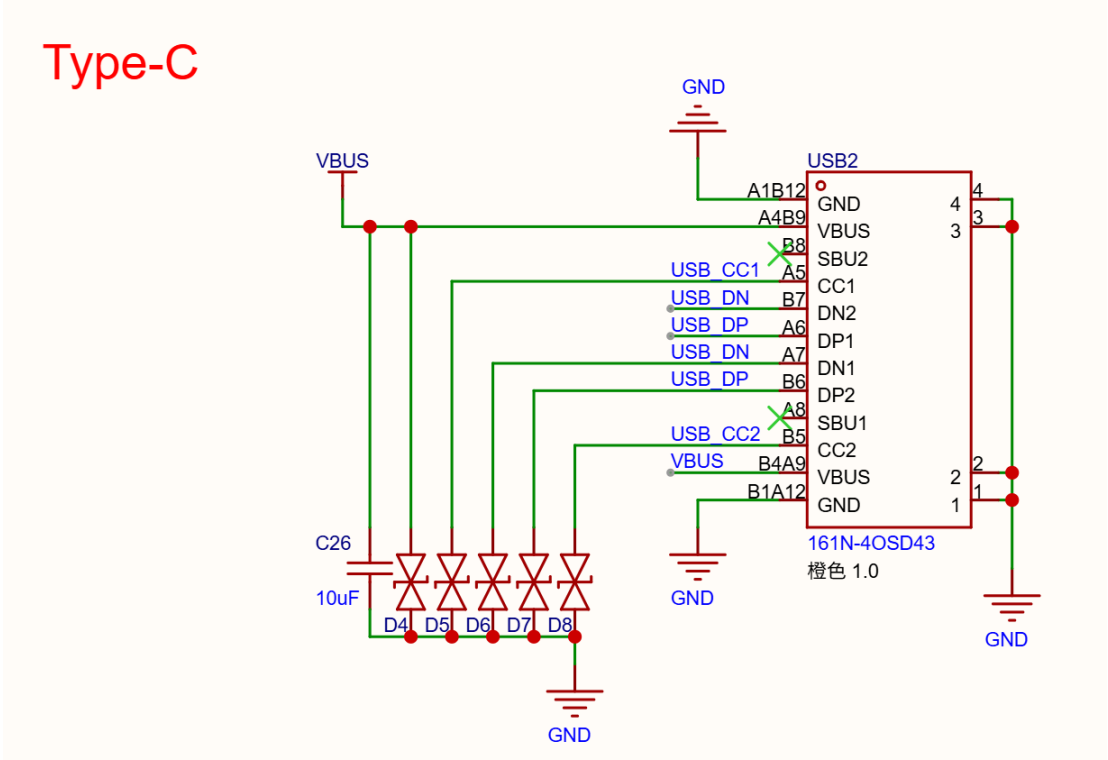

Fig. 9: Type-C Interface Schematic (Click to enlarge)

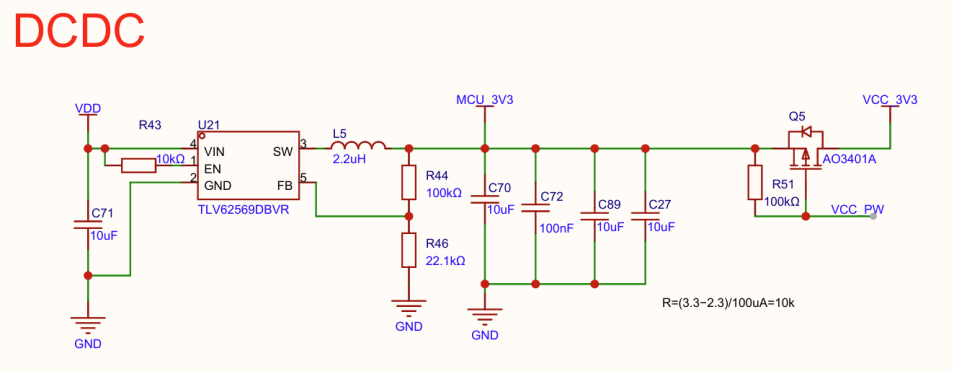

Fig. 10: DCDC 3.3 V Schematic (Click to enlarge)

Signal	GPIO	Description
VCC_3V3 / VCC_PW	GPIO60	Display supply and 3.3 V / 5 V output control, **active-low**.
POWER_SWITCH	GPIO57	Whole-device power-off request; kept high-Z during normal BSP operation, driven open-drain low to request shutdown.

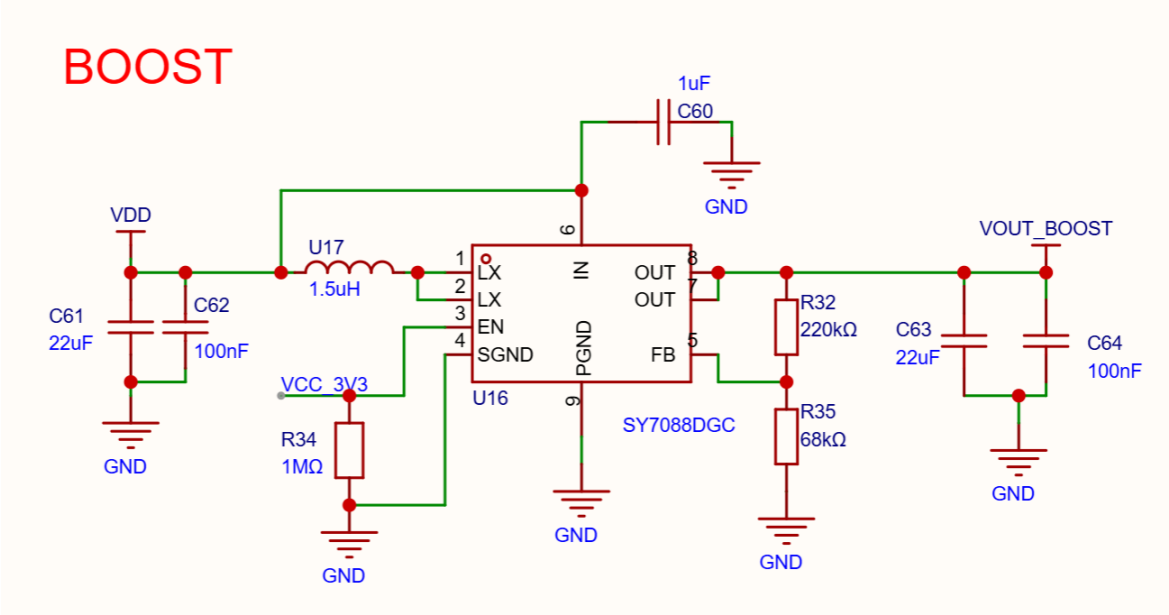

Fig. 11: BOOST Schematic (Click to enlarge)

BOOST Circuit The onboard SY7088 boost converter is enabled by `VCC_3V3` and outputs `VOUT_BOOST` for rails higher than the battery voltage.

Power On/Off Circuit SAM8108 works with the POWER button for whole-device power control. The `POWER_SWITCH` signal can also be monitored or driven by software to request shutdown.

MCU and Pin Assignment The following table lists the main pin assignments used by the ESP-Mosaico BSP, grouped by function.

Table 1: ESP-Mosaico Main Pin Assignment

Category	Signal	GPIO	Description
I2C / Sensors	I2C0_SDA	GPIO56	Internal I2C0: touch, ES8311, BMI270, BMM150, BQ27220
	I2C0_SCL	GPIO3	Internal I2C0 clock
	I2C1_SDA	GPIO0	External I2C1: module EEPROM and expansion devices
	I2C1_SCL	GPIO1	External I2C1 clock
	SENSOR_INT	GPIO2	IMU / magnetometer interrupt or signal
	TOUCH_INT	GPIO6	Touch interrupt
HMI	AI_BUTTON	GPIO7	Application button, active-low
	MOTOR	GPIO8	Vibration motor, active-high
LCD	LCD_DATA3	GPIO9	CO5300 QSPI DATA3
	LCD_DATA2	GPIO35	CO5300 QSPI DATA2
	LCD_DATA0	GPIO36	CO5300 QSPI DATA0

continues on next page

Table 1 – continued from previous page

Category	Signal	GPIO	Description
	LCD_RST	GPIO44	LCD reset
	LCD_TE	GPIO43	LCD_TE anti-tearing sync
	LCD_SCL	GPIO42	QSPI clock
	LCD_CS	GPIO50	LCD chip select
	LCD_DATA1	GPIO51	CO5300 QSPI DATA1
Audio	I2S_BCK	GPIO37	Audio bit clock
	I2S_DOUT	GPIO40	Audio data out (DAC)
	PA_CTRL	GPIO45	Amplifier enable
	I2S_WS	GPIO49	Audio word select
	I2S_DIN	GPIO52	Audio data in (ADC)
	I2S_MCLK	GPIO54	Audio master clock
Power	POWER_SWITCH	GPIO57	Power on/off request
	VCC_3V3_CTRL	GPIO60	Display supply and 3.3 V / 5 V output control, **active-low**
NAND Flash	NAND_CLK	GPIO20	SPI NAND (SD_D0)
	NAND_D	GPIO21	SPI NAND (SD_D1 / SIO0)
	NAND_Q	GPIO22	SPI NAND (SD_D2 / SIO1)
	NAND_CS	GPIO23	SPI NAND (SD_D3)
	NAND_HOLD	GPIO24	SPI NAND (SD_CLK / SIO3)
	NAND_WP	GPIO25	SPI NAND (SD_CMD / SIO2)

I2C Device Addresses V1.2 splits onboard devices and expansion modules onto two I2C buses:

- **I2C0 (internal)**: `I2C0_SDA` = GPIO56, `I2C0_SCL` = GPIO3, connecting touch, audio codec, IMU, magnetometers, and the fuel gauge.
- **I2C1 (external)**: `I2C1_SDA` = GPIO0, `I2C1_SCL` = GPIO1, connecting module EEPROM and other external devices.

Internal I2C0 (7-bit addresses)

I2C Address	Device	Description
0x11	BMM150 #2	Onboard 3-axis magnetometer
0x12	BMM150 #3	Onboard 3-axis magnetometer
0x19	ES8311	Onboard audio codec
0x55	BQ27220	Onboard fuel gauge
0x5A	CST9220	Onboard touch controller
0x69	BMI270	Onboard 6-axis IMU

External I2C1 (7-bit addresses) **Module EEPROMs are on the attached modules, not on the mainboard**, and appear only when a module with EEPROM is plugged into the corresponding slot.

I2C Address	Device	Description
0x50	Module EEPROM (Left)	On the left module, not the mainboard; selected when GPIO14 is driven low
0x51	Module EEPROM (Right)	On the right module, not the mainboard; selected when GPIO39 is driven high

LCD Interface

- LCD driver: CO5300, QSPI interface, 480 × 480 resolution.
- Touch controller: CST9220 series, I2C interface (`TP_SCL` / `TP_SDA` / `TP_RST` / `TP_INT`).

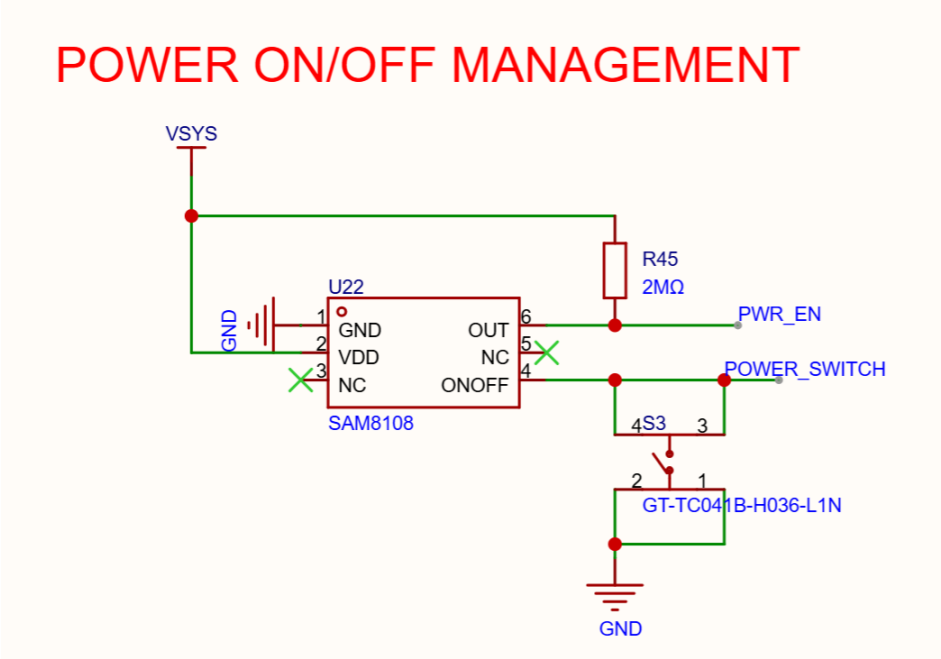

Fig. 12: Power On/Off Schematic (Click to enlarge)

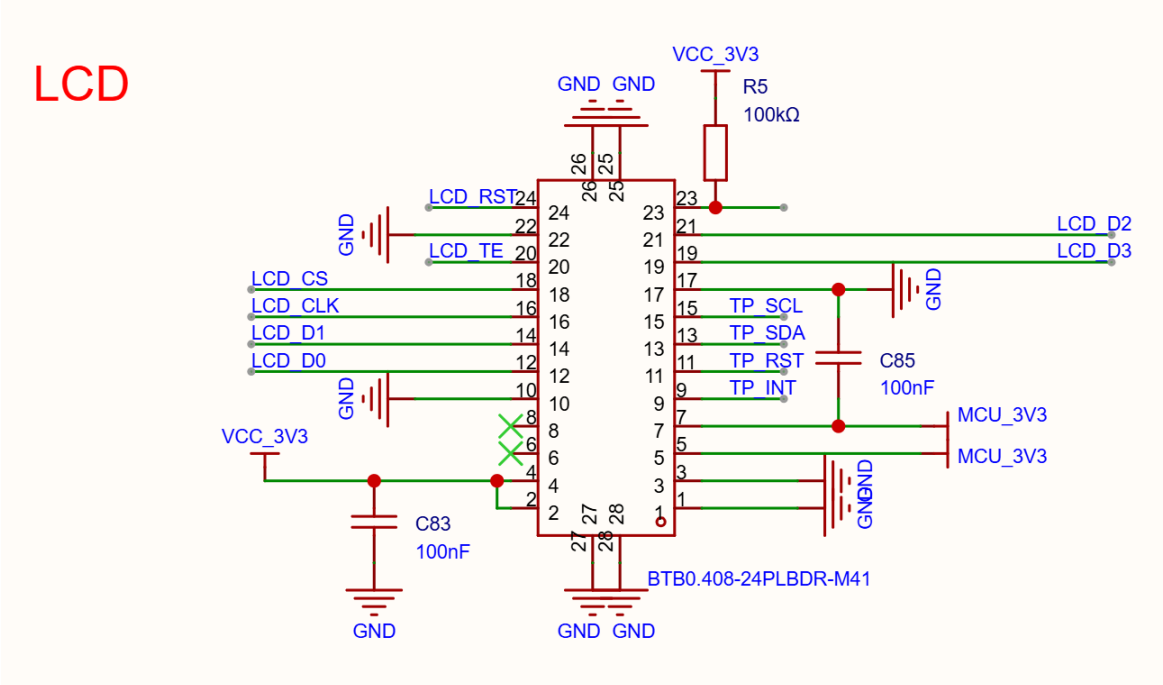

Fig. 13: LCD Interface Schematic (Click to enlarge)

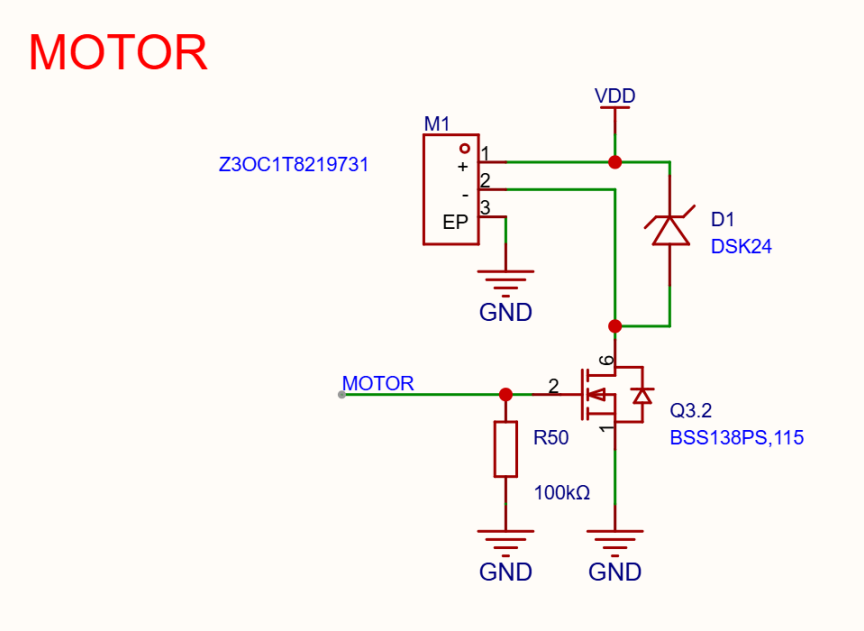

Fig. 14: Vibration Motor Schematic (Click to enlarge)

Vibration Motor Interface The vibration motor is driven by `MOTOR` (GPIO8) through a low-side MOSFET, active-high.

Module Interface The BaseBoard provides two `2 × 10P`, 2.54 mm pitch expansion headers (`H2` left and `H1` right) that expose GPIOs, UART, USB Serial/JTAG, power, and ground for secondary development. Except for power and ground, **all expansion signal pins support GPIO and can be flexibly remapped through the GPIO matrix**. Labels such as ADC / TOUCH / COMP / DAC in the tables below mark the pin's analog capabilities and do not prevent using it as a general-purpose GPIO.

ESP-Mosaico provides left and right module slots. The BSP discovers them through EEPROM address selection:

Slot	EEPROM Address	Description
Left (`H2`)	0x50	Address-select GPIO14, driven low
Right (`H1`)	0x51	Address-select GPIO39, driven high; mounted 180° relative to the left slot

Left Module Slot (`H2`) Pins Pins are listed clockwise around the connector, starting from Pin1 (odd-numbered row first, then even-numbered row back).

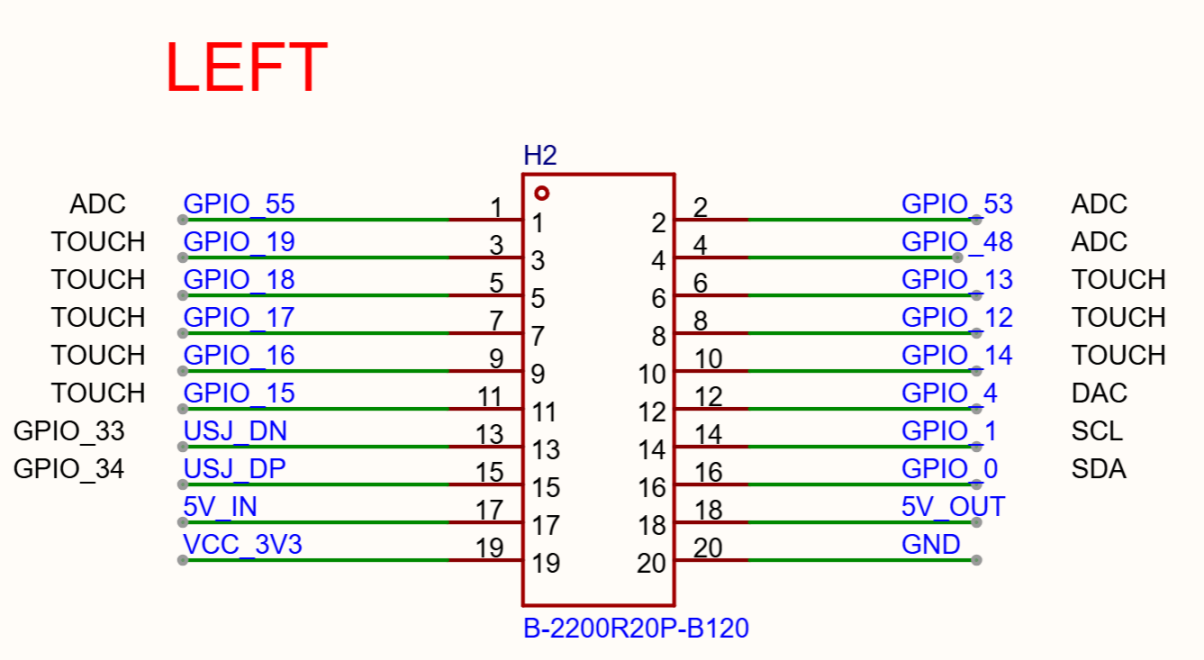

Fig. 15: Left Module Interface Schematic (Click to enlarge)

Pin	Signal	Description
1	GPIO55	ADC
3	GPIO19	ADC
5	GPIO18	TOUCH
7	GPIO17	TOUCH
9	GPIO16	TOUCH
11	GPIO15	TOUCH
13	USJ_DN (GPIO33)	USB Serial/JTAG D-
15	USJ_DP (GPIO34)	USB Serial/JTAG D+
17	5V_IN	External 5 V input (can power and charge the device)
19	VCC_3V3	3.3 V output (controlled by GPIO60)
20	GND	Ground
18	5V_OUT	5 V output (controlled by GPIO60)
16	GPIO0	SDA (external I2C1)
14	GPIO1	SCL (external I2C1)
12	GPIO4	DAC
10	GPIO14	TOUCH; EEPROM address select
8	GPIO12	TOUCH
6	GPIO13	TOUCH
4	GPIO48	ADC
2	GPIO53	ADC

Right Module Slot (`H1`) Pins Pins are listed clockwise around the connector, starting from Pin1 (odd-numbered row first, then even-numbered row back).

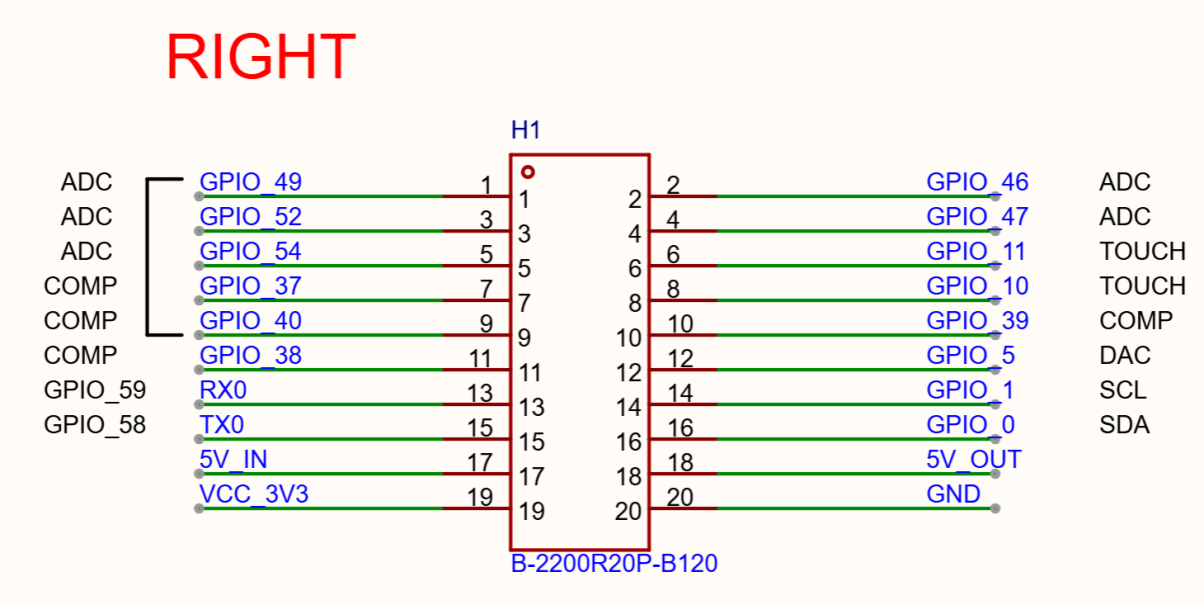

Fig. 16: Right Module Interface Schematic (Click to enlarge)

Pin	Signal	Description
1	GPIO49 *	ADC; multiplexed with codec `I2S_WS`
3	GPIO52 *	ADC; multiplexed with codec `I2S_DIN`
5	GPIO54 *	ADC; multiplexed with codec `I2S_MCLK`
7	GPIO37 *	COMP; multiplexed with codec `I2S_BCK`
9	GPIO40 *	COMP; multiplexed with codec `I2S_DOUT`
11	GPIO38	COMP
13	RX0 (GPIO59)	UART RX
15	TX0 (GPIO58)	UART TX
17	5V_IN	External 5 V input (can power and charge the device)
19	VCC_3V3	3.3 V output (controlled by GPIO60)
20	GND	Ground
18	5V_OUT	5 V output (controlled by GPIO60)
16	GPIO0	SDA (external I2C1)
14	GPIO1	SCL (external I2C1)
12	GPIO5	DAC
10	GPIO39	TOUCH; EEPROM address select
8	GPIO10	TOUCH
6	GPIO11	TOUCH
4	GPIO47	ADC
2	GPIO46	ADC

Important: Pins marked with * (**5** in total) are multiplexed with the onboard audio codec I2S bus: `GPIO49 (I2S_WS)`, `GPIO52 (I2S_DIN)`, `GPIO54 (I2S_MCLK)`, `GPIO37 (I2S_BCK)`, and `GPIO40 (I2S_DOUT)`. These pins can be used for module expansion **only when codec audio is not required**. Do not use them for expansion if the application still needs the onboard ES8311 / speaker / microphone path.

Note: When the left camera module claims DVP pins, GPIO14 (EEPROM address-select) is reused as DVP D4 and EEPROM access to that slot is suspended until the camera resource is released. See *ESP-Mosaico Camera Module* and *ESP-Mosaico Multi-function Interaction Module* for the camera module and multi-function interaction module.

Hardware Revision Details

- **V1.2** (this document): Major changes compared with V1.0 are as follows.
  - Display interface: `LCD_RST` and `LCD_SCL` are swapped (V1.2: `LCD_RST` = GPIO44, `LCD_SCL` = GPIO42).
  - The charge LED colors are changed: V1.0 uses red while charging and green when fully charged; V1.2 uses orange while charging and white when fully charged.
  - The programmable orange LED (formerly GPIO3) is removed.
  - The 3.3 V power LED is removed.
  - Independent codec power-domain control (formerly GPIO56) is removed.
  - Onboard devices and expansion modules use two I2C buses: internal I2C0 (SDA = GPIO56, SCL = GPIO3); external I2C1 remains SDA = GPIO0, SCL = GPIO1.
  - The external 3.3 V rail is generated by a dedicated DCDC, so expansion modules do not affect the internal MCU supply.
- **V1.0**: See *ESP-Mosaico V1.0*.

Related Documents

- ESP-Mosaico CoreBoard V1.2 Schematic (PDF)
- ESP-Mosaico BaseBoard V1.2 Schematic (PDF)
- 2.16-inch 480 × 480 QSPI AMOLED Display Datasheet (PDF)
- ESP32-S31 Series Datasheet (HTML)
- *Camera Module User Guide*
- *Multi-function Interaction Module User Guide*
- ESP-Mosaico User Guide (HTML)
- ESP-Mosaico Firmware Update (HTML)
- ESP-Claw (HTML)

Disclaimer and Copyright Notice

Please refer to the *Disclaimer and Copyright Notice*.

3.2.2 ESP-Mosaico V1.0

Latest version: *ESP-Mosaico V1.2*

Note: This document covers CoreBoard **V1.0** hardware. Please check the silkscreen version on the mainboard to confirm your board revision.

The camera module is supported in the **left slot only** (`H2`). Do not insert it into the right slot. For installation and usage instructions, see the *Camera Module User Guide*.

This guide helps you get started with ESP-Mosaico and provides detailed information about the board.

ESP-Mosaico is an expandable smart interaction development kit based on ESP32-S31. It targets magnetic expansion toys, motion sensing, square touch display HMI, and edge voice/multimedia applications. The device integrates a 480 × 480 QSPI square touch display, ES8311 audio codec, NS4150B amplifier, BMI270 6-axis IMU, dual BMM150 magnetometers, SPI NAND flash, and left/right 2 × 10P, 2.54 mm pitch module headers.

The main controller is Espressif ESP32-S31, supporting 2.4 GHz Wi-Fi 6 (IEEE 802.11b/g/n/ax), Bluetooth® 5.4 (LE + BR/EDR), and IEEE 802.15.4 (Thread / Zigbee). For storage, the board provides 16 MB in-package PSRAM, onboard 16 MB SPI NOR flash, and onboard 1 Gbit SPI NAND flash for graphics, multimedia, and asset caching. The display uses a CO5300-driven 480 × 480 QSPI square touch display with a CST9220-series touch controller.

For audio, ESP-Mosaico integrates ES8311 and NS4150B Class-D PA. The power system supports USB Type-C 5 V and a 3.7 V / 65 mAh Li-ion battery, with TP4057 charging, BQ27220 fuel gauge, and SAM8108 power-switch control. Both the Type-C port and the left/right module interfaces support power input and output. A USB 2.0 High-Speed OTG port supports power, debug, and communication, and the board also reserves UART expansion capability and left/right module interfaces.

Fig. 17: ESP-Mosaico Appearance (Click to enlarge)

The document consists of the following major sections:

- *Getting Started*: Overview of the board and hardware/software setup instructions.
- *Hardware Reference*: More detailed information about the board hardware.
- *Hardware Revision Details*: Hardware revision history and known issues (if any).
- *Related Documents*: Links to related documentation.
- *Disclaimer and Copyright Notice*: Link to the disclaimer and copyright notice.

Getting Started

This section briefly introduces ESP-Mosaico and describes how to flash firmware and prepare the board.

Description of Components ESP-Mosaico consists of a CoreBoard and a BaseBoard. The front views and their main components are shown first, followed by the back views and their main components.

Front The following describes the main components on the front PCB in clockwise order.

Fig. 18: ESP-Mosaico Front (Click to enlarge)

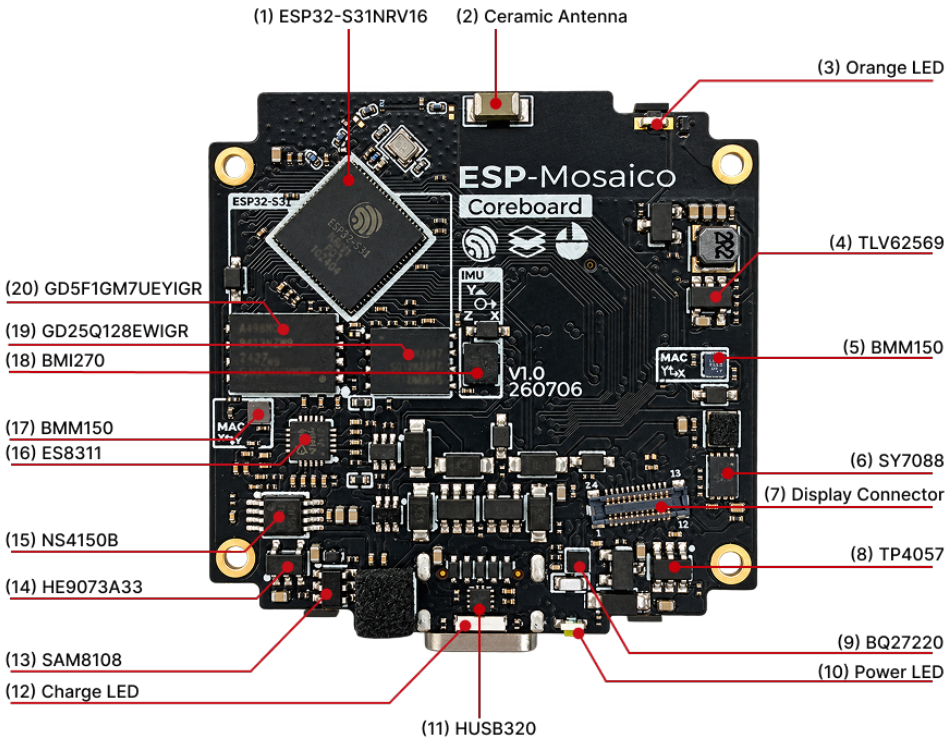

Fig. 19: ESP-Mosaico CoreBoard Front (Click to enlarge)

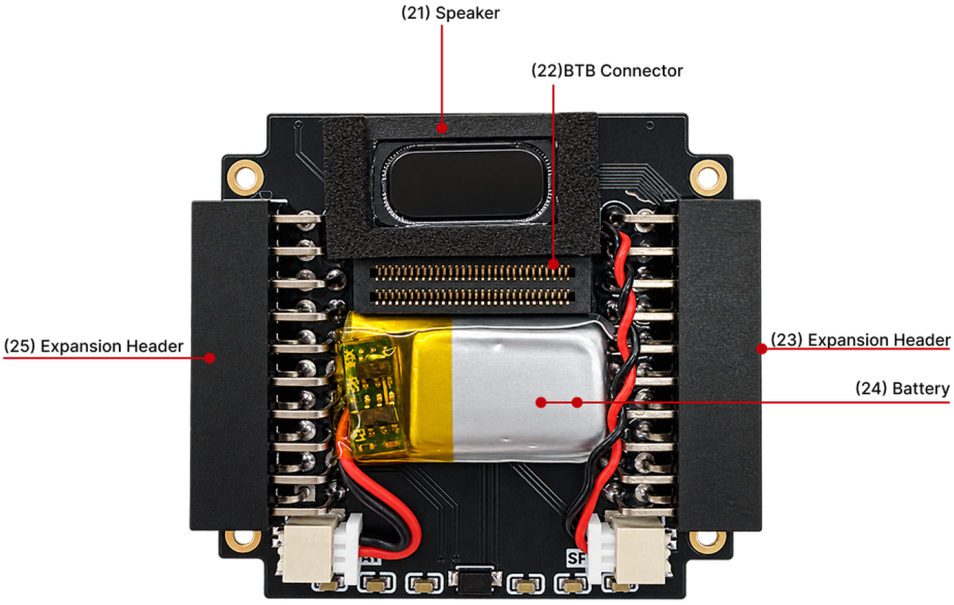

Fig. 20: ESP-Mosaico BaseBoard Front (Click to enlarge)

No.	Key Component	Description
	CoreBoard	Core board integrating the SoC, RF antenna, power management, display connector, audio codec, and sensors.
1	ESP32-S31NRV16	ESP32-S31 SoC, RISC-V 32-bit dual-core up to 320 MHz with 16 MB in-package PSRAM; supports 2.4 GHz Wi-Fi 6, Bluetooth 5.4 (LE + BR/EDR), and IEEE 802.15.4.
2	KH-3216-A35 Ceramic Antenna	2.4 GHz chip ceramic antenna for Wi-Fi and Bluetooth RF.
3	Orange LED	Programmable mono status LED on GPIO3, active-low.
4	TLV62569	Buck (DCDC) converter that converts the input supply to the system 3.3 V rail.
5	BMM150	3-axis magnetometer (BMM150 #2), I2C address 0x11.
6	SY7088	Boost (BOOST) converter for rails higher than the battery voltage.
7	Display Connector	Connects the 480 × 480, 2.16-inch QSPI touch display (CO5300 driver + CST9220 touch).
8	TP4057	Li-ion battery charger for the onboard battery.
9	BQ27220	Fuel gauge at I2C address 0x55 for voltage, current, state of charge (SOC), and related status.
10	3V3 Power LED	Lights when the 3.3 V rail is present after power-on; turns off when the device is powered off.
11	HUSB320	USB Type-C interface controller that handles CC detection and controls the USB power direction.
12	Charge LED	Charge status indicator: red while charging, green when fully charged.
13	SAM8108	Power on/off controller used with the POWER button.
14	HE9073A33	LDO providing a dedicated supply to the audio codec to reduce noise.
15	NS4150B	Low-EMI, filterless 3 W mono Class-D amplifier for the speaker.
16	ES8311	Low-power mono audio codec, I2C address 0x19, connected to the SoC over I2S.
17	BMM150	3-axis magnetometer (BMM150 #3), I2C address 0x12.
18	BMI270	6-axis IMU (accelerometer + gyroscope), I2C address 0x69.
19	GD25Q128EWIGR	16 MB (128 Mbit) SPI NOR flash for program and data storage.
20	GD5F1GM7UEYIGR	1 Gbit (128 MB) SPI NAND flash for large-capacity asset storage.
	BaseBoard	Base board integrating the speaker, battery, and left/right 2 × 10P module headers, mated to the CoreBoard through a board-to-board connector.
21	Speaker	Onboard speaker driven by the NS4150B amplifier.
22	BTB Connector	60-pin 0.5 mm board-to-board connector linking the CoreBoard and BaseBoard.
23	Expansion Header 1	2 × 10P dual-row, 2.54 mm pitch header exposing GPIOs and power for expansion.

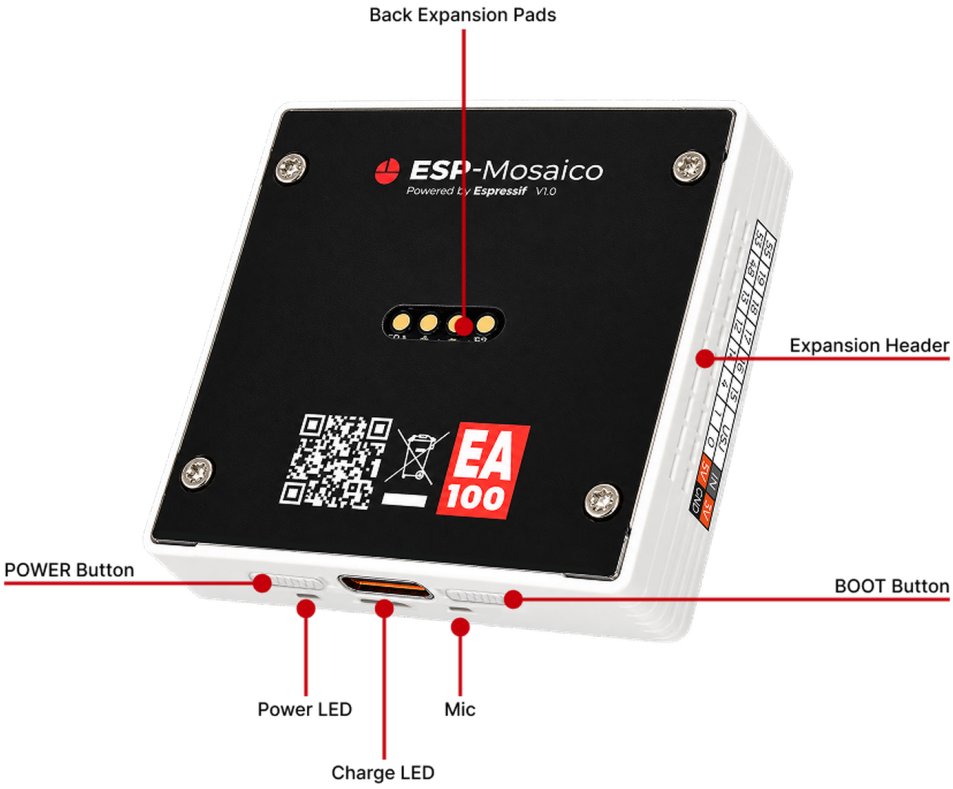

Fig. 21: ESP-Mosaico Back (Click to enlarge)

Back The following describes the main components on the back PCB in clockwise order.

No.	Key Component	Description
	CoreBoard	Core board back side, exposing the buttons, Type-C connector, microphone, and vibration motor.
1	BTB Connector	Board-to-board connector mating with the BaseBoard (back side).
2	Vibration Motor	Vibration motor controlled by GPIO8 (active-high) for haptic feedback.
3	BOOT Button	Hold during power-up to enter firmware download mode.
4	LMA2718B331-OE1	Onboard microphone connected to the ES8311 for voice capture.
5	Type-C Connector	USB 2.0 High-Speed interface for power, debug/download, and USB communication, with Li-ion charging support.
6	POWER Button	Single press toggles the device power on/off.
7	Function Button	Application button (AI button) on GPIO7, active-low.
	BaseBoard	Base board back side, providing debug pads and back expansion power pads.
8	Debug Pads	Test points exposing GND / TX / RX / BOOT / EN / 3V3 debug signals.
9	Back Expansion Pads	Back-side expansion pads exposing I2C and power interfaces for supplying the device.

Start Application Development Before powering up, make sure ESP-Mosaico is in good condition.

Required Hardware

- ESP-Mosaico
- USB cable (data-capable)
- Computer (Windows, Linux, or macOS)

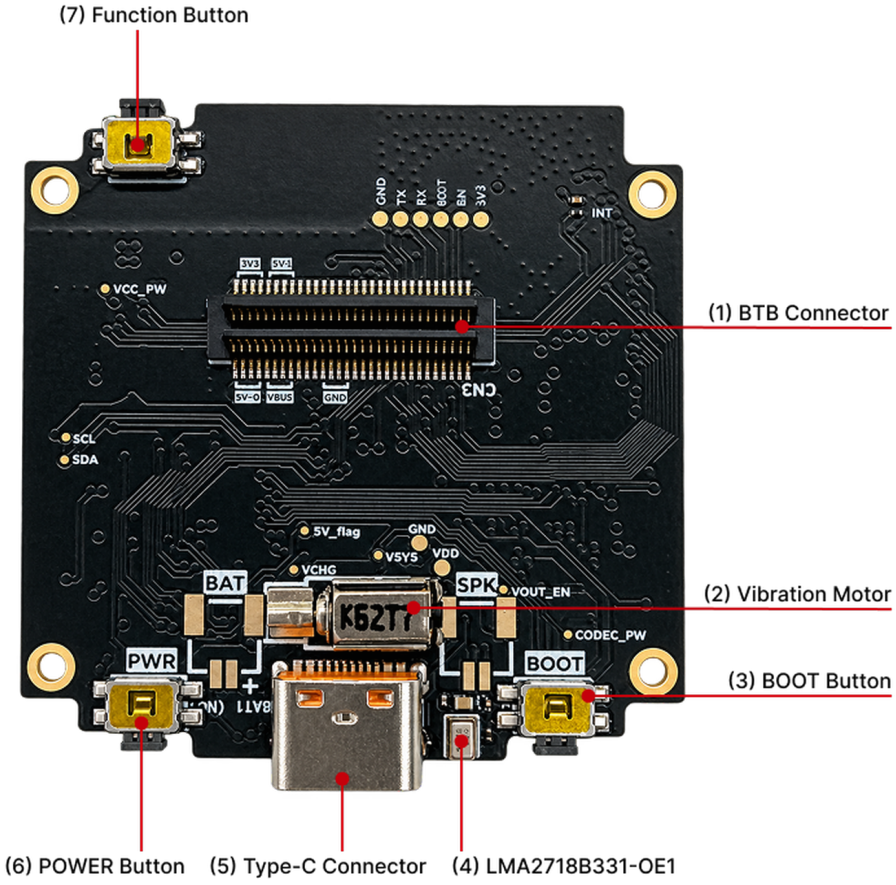

Fig. 22: ESP-Mosaico CoreBoard Back (Click to enlarge)

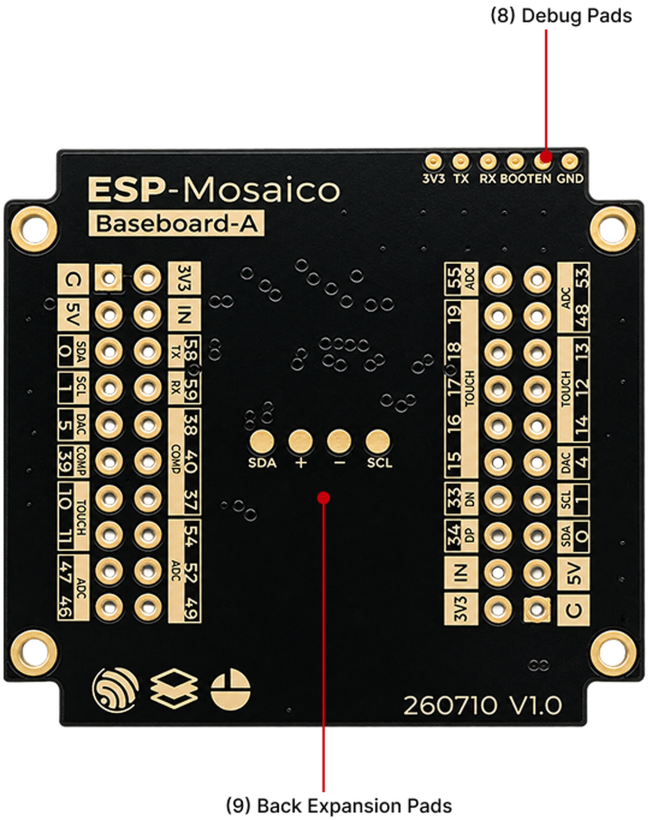

Fig. 23: ESP-Mosaico BaseBoard Back (Click to enlarge)

Note: Be sure to use a proper USB data cable. Some cables are charge-only and cannot be used for data transfer or programming.

Hardware Notes

1. Flashing
   Connect ESP-Mosaico to the computer with a USB cable through the onboard Type-C port. This port is USB 2.0 High-Speed and supports both power delivery and USB data communication. The native ESP32-S31 chip only supports firmware download through this port in download mode, and does not provide auto-download or log output by itself. The factory firmware integrates a CDC virtual serial port and supports auto-download and serial log output. If flashing fails, try holding the BOOT button while the device is powered off, then power it on to enter download mode. In addition, the left and right module interfaces provide USB Serial/JTAG (left) and a TX/RX UART (right), both of which can be used for debugging and firmware download.
2. Power
   The Type-C port can power the device. The onboard battery is charged at a rated current of 65 mA. When a USB sink device is attached, the Type-C port automatically switches to source mode, with a maximum output of 5 V / 500 mA. If the load exceeds 1 A, the port shuts off automatically and recovers after a period of time. The left and right module interfaces also provide outward 3.3 V and 5 V rails (`3V` / `5V`). These rails are output only when GPIO60 is driven low. `VIN` is an input supply pin that can also charge the onboard battery.

Software Setup Follow the ESP-IDF Get Started guide to set up the development environment. The target chip for this board is `esp32s31`.

Note: The board communicates with the PC over USB. Most operating systems (Windows, Linux, macOS) already include the required drivers. If the device is not recognized or the serial connection fails, see Establish Serial

Connection with ESP32-S31 for driver installation details.

BSP and Examples Espressif provides a board support package (BSP) for ESP-Mosaico to simplify initialization of the LCD, touch, audio, sensors, battery, NAND flash, module resources, and USB console.

- For the shipped application, buttons, expansion I/O, and getting started, see the ESP-Mosaico User Guide.
- Flash the latest firmware in the browser at Firmware Update.
- Recommended application framework: ESP-Claw.
- After the first successful flash, use the BSP USB CDC console and auto-download support. Typical commands:

```
idf.py set-target esp32s31
idf.py -p /dev/ttyACM0 flash monitor
```

Note:

- The first flash may require manually entering ROM download mode.
- The USB device briefly disconnects while the chip resets; IDF Monitor usually reconnects automatically.
- Use `monitor --no-reset` when attaching without restarting the application.
- Related options: `CONFIG_BSP_USB_CONSOLE`, `CONFIG_BSP_USB_CONSOLE_AUTO_INIT`, `CONFIG_BSP_USB_AUTO_DOWNLOAD`.

Hardware Reference

Block Diagram The main components and connections of ESP-Mosaico are shown below.

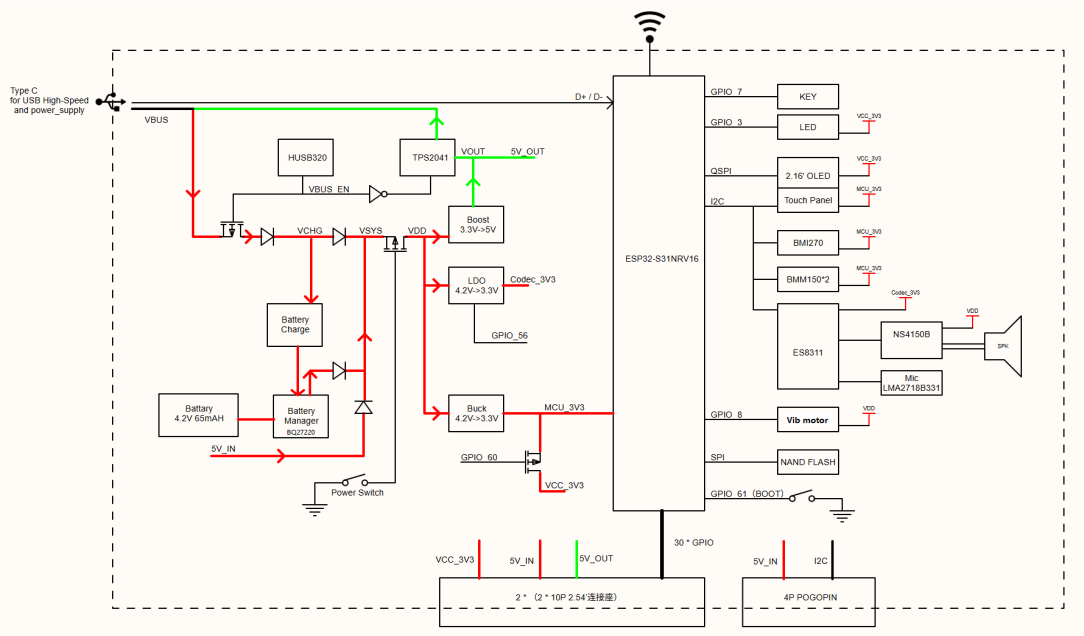

Fig. 24: ESP-Mosaico Block Diagram (Click to enlarge)

Power Supply Options You can power the board in the following ways:

1. Through the `Type-C (USB-C)` port
   Connect a USB Type-C cable to the Type-C port.
2. Through the `back expansion pads`
   Supply external power through the back expansion pads on the BaseBoard.

3. Through the `battery`
   The board supports a 3.7 V / 65 mAh Li-ion battery.

Regardless of the power source, use the POWER button to turn the device on or off. External power sources above can also charge the onboard battery.

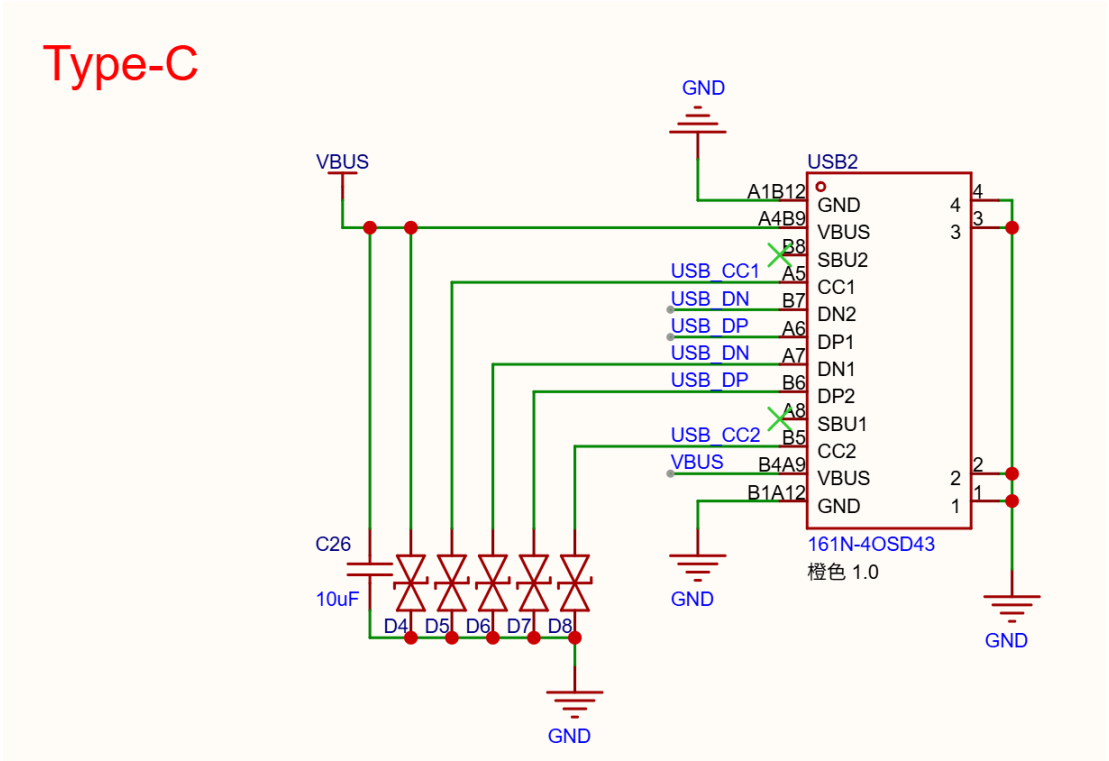

Fig. 25: Type-C Interface Schematic (Click to enlarge)

Type-C Interface The Type-C port connects to ESP32-S31 USB 2.0 High-Speed OTG for power, debug, and application communication. Onboard USB Serial/JTAG signals are used for download and debugging. The BSP USB CDC console and auto-download feature are the recommended download and log path.

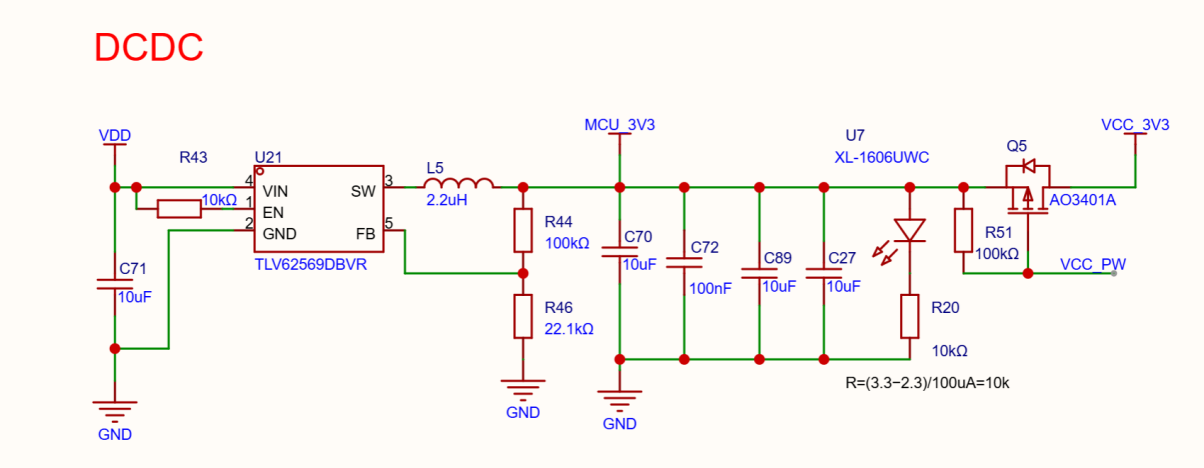

Fig. 26: DCDC 3.3 V Schematic (Click to enlarge)

DCDC 3.3 V Circuit TLV62569 steps `VDD` down to `MCU_3V3`, which is then switched to `VCC_3V3` by the `VCC_PW` load switch. `U7` is the 3.3 V power LED.

Power-control GPIOs defined by the BSP:

Signal	GPIO	Description
VCC_3V3 / VCC_PW	GPIO60	System 3.3 V rail control, **active-low**; BSP applies a soft-start ramp when enabling.
CODEC_3V3	GPIO56	Codec power control, **active-high**.
POWER_SWITCH	GPIO57	Whole-device power-off request; kept high-Z during normal BSP operation, driven open-drain low to request shutdown.

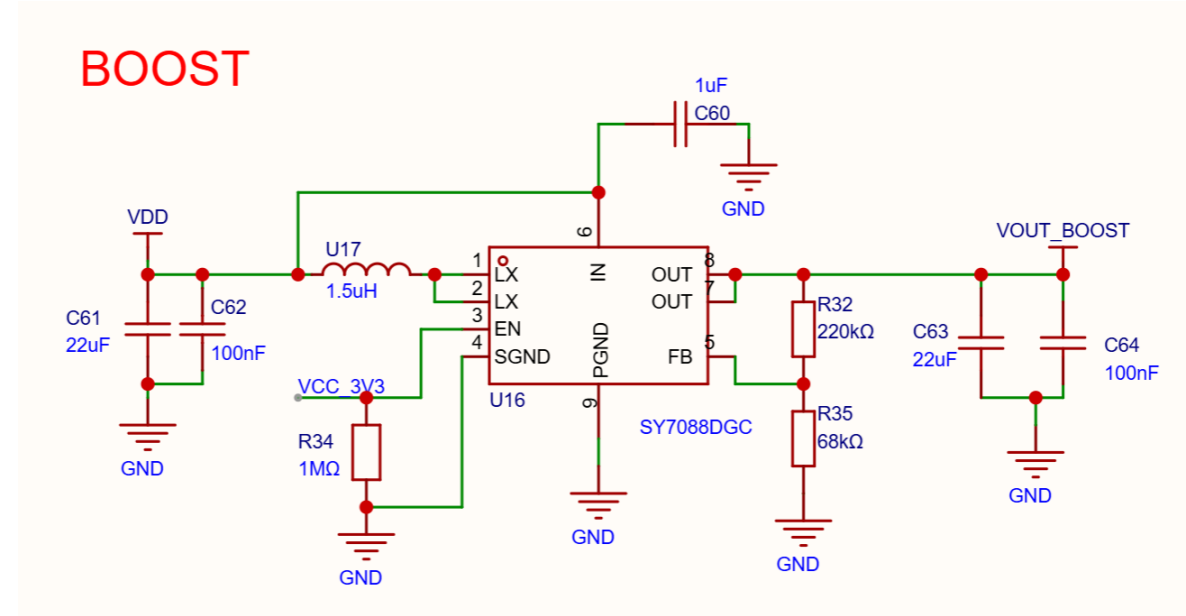

Fig. 27: BOOST Schematic (Click to enlarge)

BOOST Circuit The onboard SY7088 boost converter is enabled by `VCC_3V3` and outputs `VOUT_BOOST` for rails higher than the battery voltage.

Power On/Off Circuit SAM8108 works with the POWER button for whole-device power control. The `POWER_SWITCH` signal can also be monitored or driven by software to request shutdown.

MCU and Pin Assignment The following table lists the main pin assignments used by the ESP-Mosaico BSP, grouped by function.

Table 2: ESP-Mosaico Main Pin Assignment

Category	Signal	GPIO	Description
I2C / Sensors	I2C0_SDA	GPIO0	Shared I2C for touch, ES8311, BMI270, BMM150, BQ27220, and module EEPROM
	I2C0_SCL	GPIO1	Shared I2C clock
	SENSOR_INT	GPIO2	IMU / magnetometer interrupt or signal
	TOUCH_INT	GPIO6	Touch interrupt
HMI	STATUS_LED	GPIO3	Orange status LED, software-controlled, active-low
	AI_BUTTON	GPIO7	Application button, active-low
	MOTOR	GPIO8	Vibration motor, active-high
LCD	LCD_DATA3	GPIO9	CO5300 QSPI DATA3

continues on next page

Table 2 – continued from previous page

Category	Signal	GPIO	Description
	LCD_DATA2	GPIO35	CO5300 QSPI DATA2
	LCD_DATA0	GPIO36	CO5300 QSPI DATA0
	LCD_RST	GPIO42	LCD reset
	LCD_TE	GPIO43	LCD_TE anti-tearing sync
	LCD_SCL	GPIO44	QSPI clock
	LCD_CS	GPIO50	LCD chip select
	LCD_DATA1	GPIO51	CO5300 QSPI DATA1
Audio	I2S_BCK	GPIO37	Audio bit clock
	I2S_DOUT	GPIO40	Audio data out (DAC)
	PA_CTRL	GPIO45	Amplifier enable
	I2S_WS	GPIO49	Audio word select
	I2S_DIN	GPIO52	Audio data in (ADC)
	I2S_MCLK	GPIO54	Audio master clock
	CODEC_PW	GPIO56	Codec 3.3 V power control
Power	POWER_SWITCH	GPIO57	Power on/off request
	VCC_3V3_CTRL	GPIO60	System 3.3 V power control
NAND Flash	NAND_CLK	GPIO20	SPI NAND (SD_D0)
	NAND_D	GPIO21	SPI NAND (SD_D1 / SIO0)
	NAND_Q	GPIO22	SPI NAND (SD_D2 / SIO1)
	NAND_CS	GPIO23	SPI NAND (SD_D3)
	NAND_HOLD	GPIO24	SPI NAND (SD_CLK / SIO3)
	NAND_WP	GPIO25	SPI NAND (SD_CMD / SIO2)

I2C Device Addresses The 7-bit addresses on the shared I2C bus (`I2C0_SDA` / `I2C0_SCL`) are listed below. Onboard devices are on the CoreBoard / BaseBoard. **Module EEPROMs are on the attached modules, not on the mainboard**, and appear only when a module with EEPROM is plugged into the corresponding slot.

I2C Address	Device	Description
0x11	BMM150 #2	Onboard 3-axis magnetometer
0x12	BMM150 #3	Onboard 3-axis magnetometer
0x19	ES8311	Onboard audio codec
0x50	Module EEPROM (Left)	On the left module, not the mainboard; selected when GPIO14 is driven low
0x51	Module EEPROM (Right)	On the right module, not the mainboard; selected when GPIO39 is driven high
0x55	BQ27220	Onboard fuel gauge
0x5A	CST9220	Onboard touch controller
0x69	BMI270	Onboard 6-axis IMU

LCD Interface

- LCD driver: CO5300, QSPI interface, 480 × 480 resolution.
- Touch controller: CST9220 series, I2C interface (`TP_SCL` / `TP_SDA` / `TP_RST` / `TP_INT`).

Vibration Motor Interface The vibration motor is driven by `MOTOR` (GPIO8) through a low-side MOSFET, active-high.

Module Interface The BaseBoard provides two `2 × 10P`, 2.54 mm pitch expansion headers (`H2` left and `H1` right) that expose GPIOs, UART, USB Serial/JTAG, power, and ground for secondary development. Except for power and ground, **all expansion signal pins support GPIO and can be flexibly remapped through the GPIO**

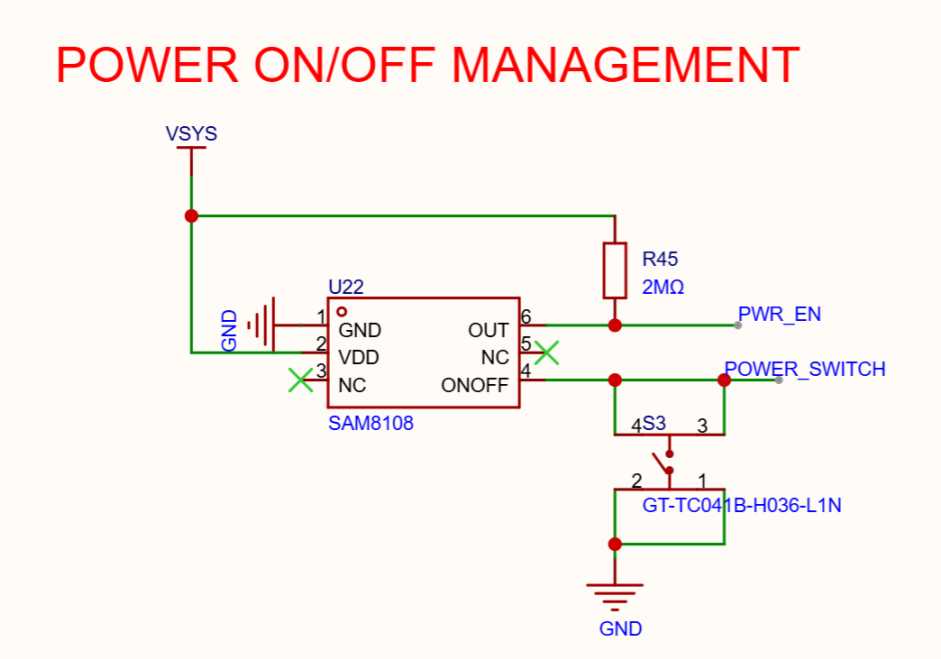

Fig. 28: Power On/Off Schematic (Click to enlarge)

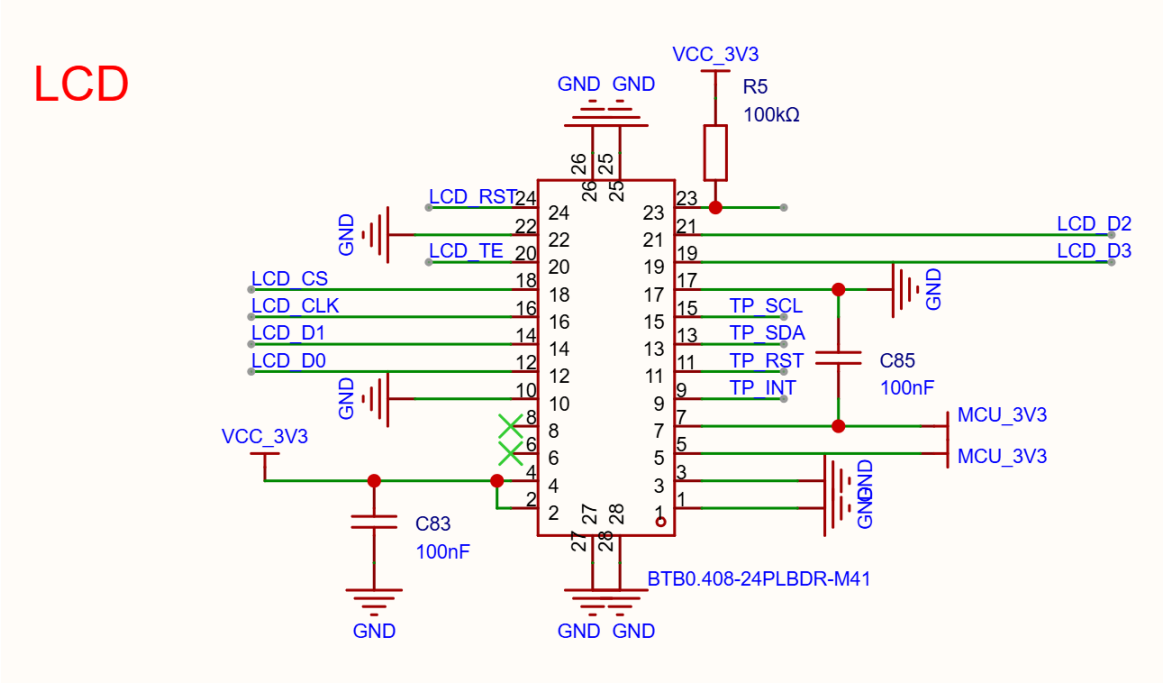

Fig. 29: LCD Interface Schematic (Click to enlarge)

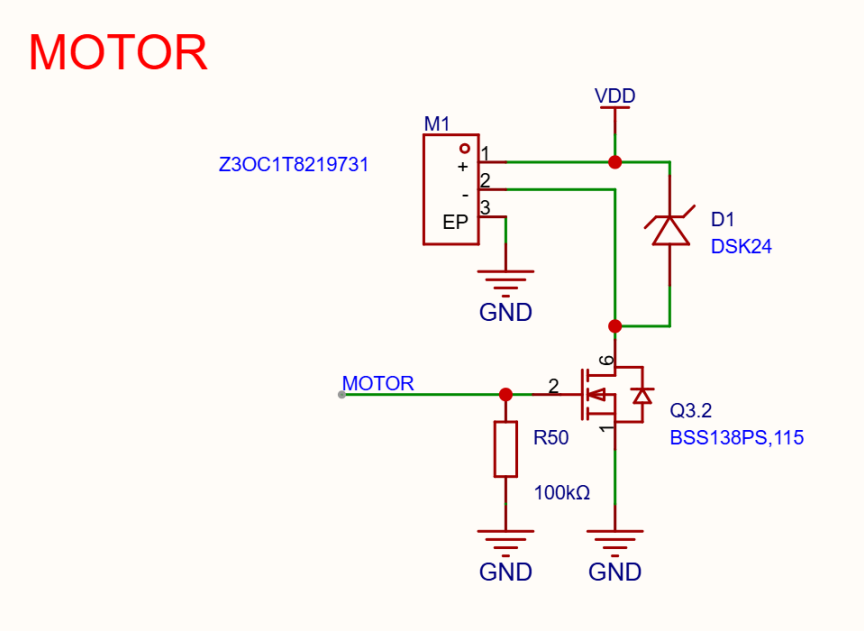

Fig. 30: Vibration Motor Schematic (Click to enlarge)

matrix. Labels such as ADC / TOUCH / COMP / DAC in the tables below mark the pin ' s analog capabilities and do not prevent using it as a general-purpose GPIO.

ESP-Mosaico provides left and right module slots. The BSP discovers them through EEPROM address selection:

Slot	EEPROM Address	Description
Left (H2)	0x50	Address-select GPIO14, driven low
Right (H1)	0x51	Address-select GPIO39, driven high; mounted 180° relative to the left slot

Left Module Slot (H2) Pins Pins are listed clockwise around the connector, starting from Pin1 (odd-numbered row first, then even-numbered row back).

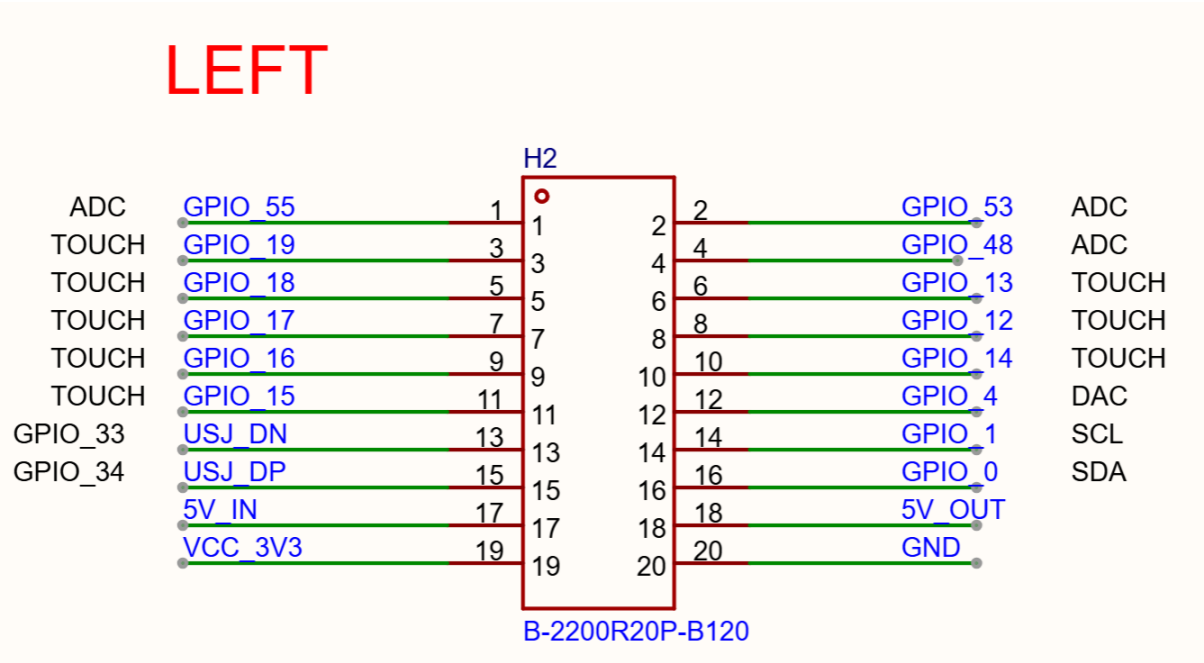

Fig. 31: Left Module Interface Schematic (Click to enlarge)

Pin	Signal	Description
1	GPIO55	ADC
3	GPIO19	ADC
5	GPIO18	TOUCH
7	GPIO17	TOUCH
9	GPIO16	TOUCH
11	GPIO15	TOUCH
13	USJ_DN (GPIO33)	USB Serial/JTAG D-
15	USJ_DP (GPIO34)	USB Serial/JTAG D+
17	5V_IN	External 5 V input (can power and charge the device)
19	VCC_3V3	3.3 V output (controlled by GPIO60)
20	GND	Ground
18	5V_OUT	5 V output (controlled by GPIO60)
16	GPIO0	SDA (shared onboard I2C)
14	GPIO1	SCL (shared onboard I2C)
12	GPIO4	DAC
10	GPIO14	TOUCH; EEPROM address select
8	GPIO12	TOUCH
6	GPIO13	TOUCH
4	GPIO48	ADC
2	GPIO53	ADC

Right Module Slot (`H1`) Pins Pins are listed clockwise around the connector, starting from Pin1 (odd-numbered row first, then even-numbered row back).

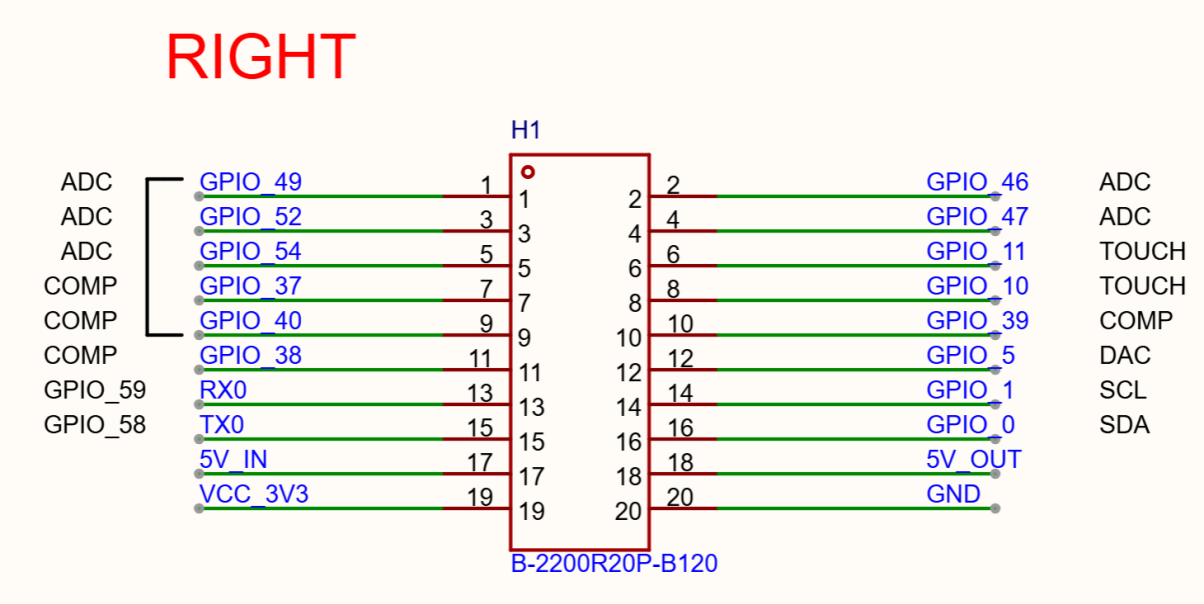

Fig. 32: Right Module Interface Schematic (Click to enlarge)

Pin	Signal	Description
1	GPIO49 *	ADC; multiplexed with codec `I2S_WS`
3	GPIO52 *	ADC; multiplexed with codec `I2S_DIN`
5	GPIO54 *	ADC; multiplexed with codec `I2S_MCLK`
7	GPIO37 *	COMP; multiplexed with codec `I2S_BCK`
9	GPIO40 *	COMP; multiplexed with codec `I2S_DOUT`
11	GPIO38	COMP
13	RX0 (GPIO59)	UART RX
15	TX0 (GPIO58)	UART TX
17	5V_IN	External 5 V input (can power and charge the device)
19	VCC_3V3	3.3 V output (controlled by GPIO60)
20	GND	Ground
18	5V_OUT	5 V output (controlled by GPIO60)
16	GPIO0	SDA (shared onboard I2C)
14	GPIO1	SCL (shared onboard I2C)
12	GPIO5	DAC
10	GPIO39	TOUCH; EEPROM address select
8	GPIO10	TOUCH
6	GPIO11	TOUCH
4	GPIO47	ADC
2	GPIO46	ADC

Important: Pins marked with * (**5** in total) are multiplexed with the onboard audio codec I2S bus: `GPIO49 (I2S_WS)`, `GPIO52 (I2S_DIN)`, `GPIO54 (I2S_MCLK)`, `GPIO37 (I2S_BCK)`, and `GPIO40 (I2S_DOUT)`. These pins can be used for module expansion **only when codec audio is not required**. Do not use them for expansion if the application still needs the onboard ES8311 / speaker / microphone path.

Note: When the left camera module claims DVP pins, GPIO14 (EEPROM address-select) is reused as DVP D4 and EEPROM access to that slot is suspended until the camera resource is released. See *ESP-Mosaico Camera Module* and *ESP-Mosaico Multi-function Interaction Module* for the camera module and multi-function interaction module.

Hardware Revision Details

- **V1.0** (this document): Initial hardware. Onboard devices and expansion modules share one I2C bus. The board has a programmable orange LED. The codec power domain can be controlled independently by GPIO. `LCD_RST` is GPIO42 and `LCD_SCL` is GPIO44. The charge LED is red while charging and green when fully charged.
- **V1.2**: `LCD_RST` and `LCD_SCL` are swapped; the charge LED is orange while charging and white when fully charged; the programmable LED and 3.3 V power LED are removed; independent codec power-domain control is removed; internal I2C0 and external I2C1 are split; the external 3.3 V rail uses a dedicated DCDC. See *ESP-Mosaico V1.2*.

Related Documents

- ESP-Mosaico CoreBoard V1.0 Schematic (PDF)
- ESP-Mosaico BaseBoard V1.0 Schematic (PDF)
- 2.16-inch 480 × 480 QSPI AMOLED Display Datasheet (PDF)
- ESP32-S31 Series Datasheet (HTML)
- *Camera Module User Guide*
- *Multi-function Interaction Module User Guide*
- ESP-Mosaico User Guide (HTML)
- ESP-Mosaico Firmware Update (HTML)
- ESP-Claw (HTML)

Disclaimer and Copyright Notice

Please refer to the *Disclaimer and Copyright Notice*.

3.2.3 ESP-Mosaico Camera Module

This guide describes the hardware versions, interfaces, installation, and software usage of the ESP-Mosaico **Camera Module** (Camera Module). The module is available in **V1.2** and **V1.4** versions. Both connect to the left `2 × 10P` slot on the CoreBoard / BaseBoard and provide DVP image capture.

Note: The camera module is supported in the **left slot only** (`H2`, EEPROM address 0x50). Do not insert it into the right slot. For the mainboard revision, see *ESP-Mosaico V1.2* or *ESP-Mosaico V1.0*.

Both versions are powered from the host expansion header, output 8-bit DVP image data, and use SCCB / I2C to communicate with the image sensor and module EEPROM. They share the same pin assignments but differ in image sensor, oscillator frequency, default output resolution, and EEPROM model and package.

This document consists of the following major sections:

- *Hardware Versions*: PCB views, components, and camera specifications for V1.2, followed by V1.4.
- *Getting Started*: Module overview, required hardware, installation, and software examples.
- *Hardware Reference*: DVP pins, power, and usage limits.
- *Related Documents*: Links to related documentation.
- *Disclaimer and Copyright Notice*: Disclaimer and copyright notice.

Hardware Versions

Identify the hardware version from the version marking on the front of the camera module PCB, then follow the corresponding link below for detailed information:

- *V1.2 Version*: Equipped with an OV3640 camera.

- *V1.4 Version*: Equipped with an SC101IOT camera.

These version numbers refer to the camera module PCB, not the CoreBoard mainboard. Each version is described in the following order: version overview, PCB front markings and interfaces, PCB back components, and camera specifications. Both versions share the installation procedure and pin assignments described in *Getting Started* and *Hardware Reference*.

V1.2 Version

V1.2 Version Overview V1.2 features an OV3640 DVP image sensor. A 24 MHz oscillator external to the sensor provides `XCLK` without using a host GPIO. The default capture format is **1024 × 768 UYVY**.

The module uses an **AT24C02BS(GMIC)** EEPROM in an **SOP-8** package.

V1.2 PCB Front Markings and Interfaces The front of the PCB has no components and mainly carries the module name, PCB version V1.2, and date markings.

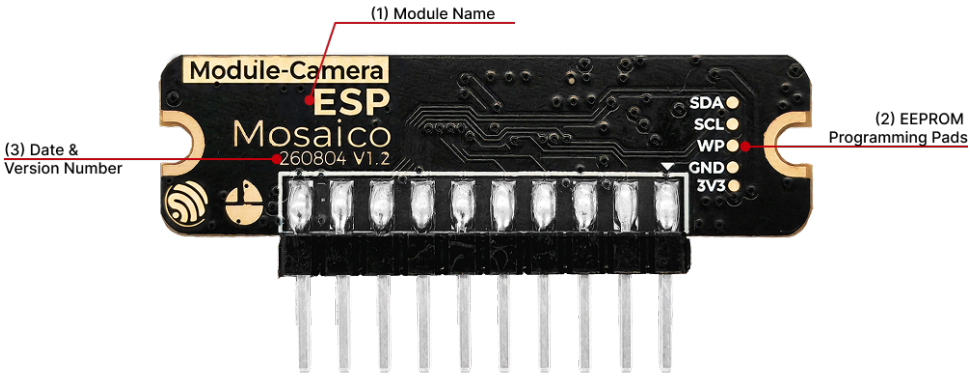

Fig. 33: V1.2 Camera Module PCB Front (Click to enlarge)

The front mainly contains silkscreen markings and test pads, as described below.

Table 3: V1.2 PCB Front Markings and Interfaces

No.	Key Component	Description
1	Module Name Markings	Marked `Module-Camera` and `ESP Mosaico` to identify the camera module.
2	Version and Date Markings	Marked `260804 V1.2` to indicate the PCB date and version.
3	EEPROM Test Pads	Labeled `SDA`, `SCL`, `WP`, `GND`, and `3V3` for data, clock, write protection, ground, and 3.3 V power, respectively.

V1.2 PCB Back Components All components are located on the back of the PCB, including the camera FPC connector, 24 MHz oscillator, two flash LEDs, camera status indicator (red LED), AT24C02BS(GMIC) EEPROM (SOP-8 package), and power supply circuitry.

The pin header that connects to ESP-Mosaico straddles the PCB edge, with solder joints on both sides of the PCB.

The main components on the back are listed below. Positions refer to the orientation in the image.

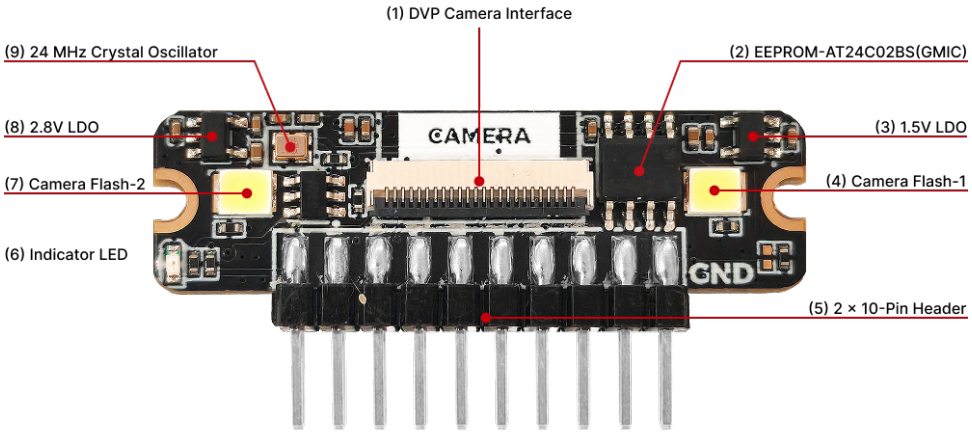

Fig. 34: V1.2 Camera Module PCB Back (Click to enlarge)

Table 4: V1.2 PCB Back Components

No.	Key Component	Description
1	Camera FPC Connector	Located in the center. Connects the OV3640 camera and carries image data, control signals, and power.
2	AT24C02BS(GMIC) EEPROM	Located to the right of the FPC connector, in an SOP-8 package. Stores module identification information; the 7-bit address in the left slot is 0x50.
3	1.5 V LDO	Located at the upper right. Supplies 1.5 V to the camera.
4	Camera Flash-1	Located on the right for supplemental lighting. Controlled by GPIO34: low turns it on and high turns it off. Configure GPIO34 as an open-drain output to prevent a brief flash when the host powers down.
5	2 × 10-Pin Header (Module Connector)	Located at the bottom. Straddles the PCB edge with solder joints on both sides, and connects to the BaseBoard left slot `H2`.
6	Camera Status Indicator (Red LED)	Located at the lower left. Lights up when the camera starts operating and turns off when the camera enters sleep or is shut down.
7	Camera Flash-2	Located on the left for supplemental lighting. Shares GPIO34 with Flash-1, with the same control levels and open-drain output requirement.
8	2.8 V LDO	Located at the upper left. Supplies 2.8 V to the camera.
9	24 MHz Oscillator	Located above and to the left of the FPC connector. Provides `XCLK` for the OV3640 without using a host GPIO.

V1.2 Camera Specifications

Table 5: V1.2 Camera Specifications

Parameter	Specification
Module PCB Version	V1.2
Image Sensor	OV3640
Sensor Type and Pixel Count	3.2-megapixel CMOS
Sensor Size	1/4 inch
Pixel Size	1.75 µm
Lens Focal Length	4.15 mm, fixed focus
Aperture	F2.4
Field of View	68°
Applications	General-purpose image capture
Lens Distortion	<1%
Default Output Format	1024 × 768 UYVY
Output Resolution and Frame Rate	1024 × 768 @ 30 FPS
Supported Image Formats	RAW / YUV / RGB
Automatic Image Controls	AEC (automatic exposure control) / AGC (automatic gain control) / AWB (automatic white balance)

V1.4 Version

V1.4 Version Overview V1.4 features an SC101IOT DVP image sensor. An onboard 20 MHz oscillator provides `XCLK` without using a host GPIO. The default output format is **1280 × 720 UYVY**.

The module uses a **24C02S(XBLW)** EEPROM in an **SOT23-5** package.

V1.4 PCB Front Markings and Interfaces The front of the PCB has no components and mainly carries the module name, PCB version V1.4, and date markings.

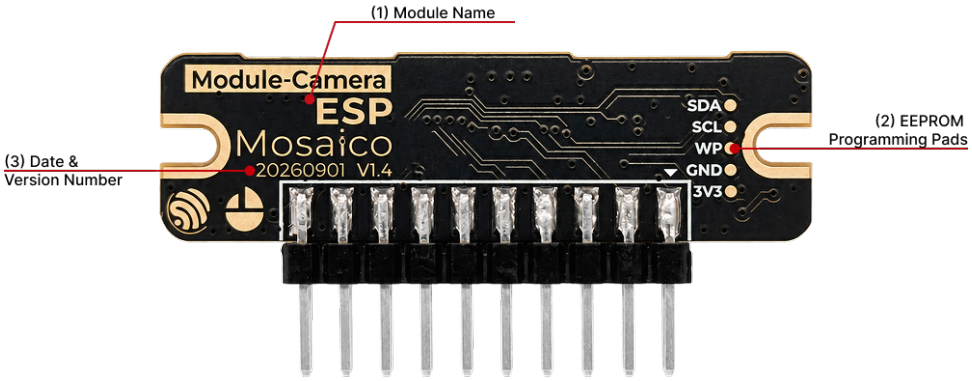

Fig. 35: V1.4 Camera Module PCB Front (Click to enlarge)

The front mainly contains silkscreen markings and test pads, as described below.

Table 6: V1.4 PCB Front Markings and Interfaces

No.	Key Component	Description
1	Module Name Markings	Marked `Module-Camera` and `ESP Mosaico` to identify the camera module.
2	Version and Date Markings	Marked `20260901 V1.4` to indicate the PCB date and version.
3	EEPROM Test Pads	Labeled `SDA`, `SCL`, `WP`, `GND`, and `3V3` for data, clock, write protection, ground, and 3.3 V power, respectively.

V1.4 PCB Back Components All components are located on the back of the PCB, including the camera FPC connector, 20 MHz oscillator, two flash LEDs, camera status indicator (red LED), 24C02S(XBLW) EEPROM (SOT23-5 package), and power supply circuitry.

The pin header that connects to ESP-Mosaico straddles the PCB edge, with solder joints on both sides of the PCB.

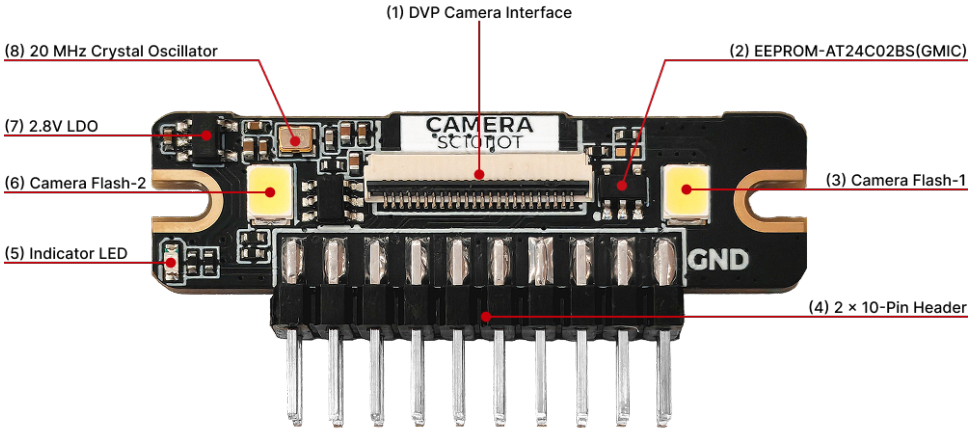

Fig. 36: V1.4 Camera Module PCB Back (Click to enlarge)

The main components on the back are listed below. Positions refer to the orientation in the image.

Table 7: V1.4 PCB Back Components

No.	Key Component	Description
1	Camera FPC Connector	Located in the center. Connects the SC101IOT camera and carries image data, control signals, and power.
2	24C02S(XBLW) EEPROM	Located to the right of the FPC connector, in an SOT23-5 package. Stores module identification information; the 7-bit address in the left slot is 0x50.
3	Camera Flash-1	Located on the right for supplemental lighting. Controlled by GPIO34: low turns it on and high turns it off. Configure GPIO34 as an open-drain output to prevent a brief flash when the host powers down.
4	2 × 10-Pin Header (Module Connector)	Located at the bottom. Straddles the PCB edge with solder joints on both sides, and connects to the BaseBoard left slot `H2`.
5	Camera Status Indicator (Red LED)	Located at the lower left. Lights up when the camera starts operating and turns off when the camera enters sleep or is shut down.
6	Camera Flash-2	Located on the left for supplemental lighting. Shares GPIO34 with Flash-1, with the same control levels and open-drain output requirement.
7	2.8 V LDO	Located at the upper left. Supplies 2.8 V to the camera.
8	20 MHz Oscillator	Located above and to the left of the FPC connector. Provides `XCLK` for the SC101IOT without using a host GPIO.

V1.4 Camera Specifications The SC101IOT camera module for V1.4 is available with a low-profile lens or a high-profile lens. Both support manual focus. The optical specifications for each lens are listed below; see the corresponding module specifications in *Related Documents* for details.

Table 8: V1.4 Camera Specifications

Parameter	Specification
Module PCB Version	V1.4
Image Sensor	SC101IOT
Sensor Type and Pixel Count	1-megapixel CMOS
Sensor Size	1/4.2 inch
Pixel Size	2.9 μm
Lens Focal Length	Low-profile lens: 3.1 mm, manual focus High-profile lens: 4.3 mm, manual focus
Aperture	Low-profile lens: F1.8 (±5%) High-profile lens: F2.4 (±5%)
Field of View	Low-profile lens: diagonal (D): 79°, horizontal (H): 73°, vertical (V): 42° High-profile lens: diagonal (D): 52°, horizontal (H): 46°, vertical (V): 26°
Applications	General-purpose image capture
Lens Distortion	Low-profile lens: <1.0% High-profile lens: <0.5%
Default Output Format	1280 × 720 UYVY
Output Resolution and Frame Rate	1280 × 720 @ 30 FPS (8-bit, maximum transfer specification)
Supported Image Formats	RAW / YUV422 / RGB
Automatic Image Controls	AEC (automatic exposure control) / AGC (automatic gain control) / AWB (automatic white balance)

Getting Started

This section applies to V1.2 and V1.4. For the PCB views, components, and specifications of each version, see *Hardware Versions*.

Module Overview The camera module has a flip design. Folding or unfolding the section that holds the lens switches between front-facing and rear-facing camera modes.

In the image below, the left column shows front-facing mode and the right column shows rear-facing mode. The upper and lower views in each column show the two sides of the module in that mode.

- **Front-facing mode (folded, left column)**: The lens section folds against the module body for a compact form. The upper-left view shows the lens side and the lower-left view shows the flash side. The lens and the two flash LEDs face opposite directions.
- **Rear-facing mode (unfolded, right column)**: The lens section unfolds so that the lens and both flash LEDs face the same direction, allowing the LEDs to provide supplemental lighting. The upper-right view shows the lens and flash LEDs; the lower-right view shows the flip mechanism on the other side.

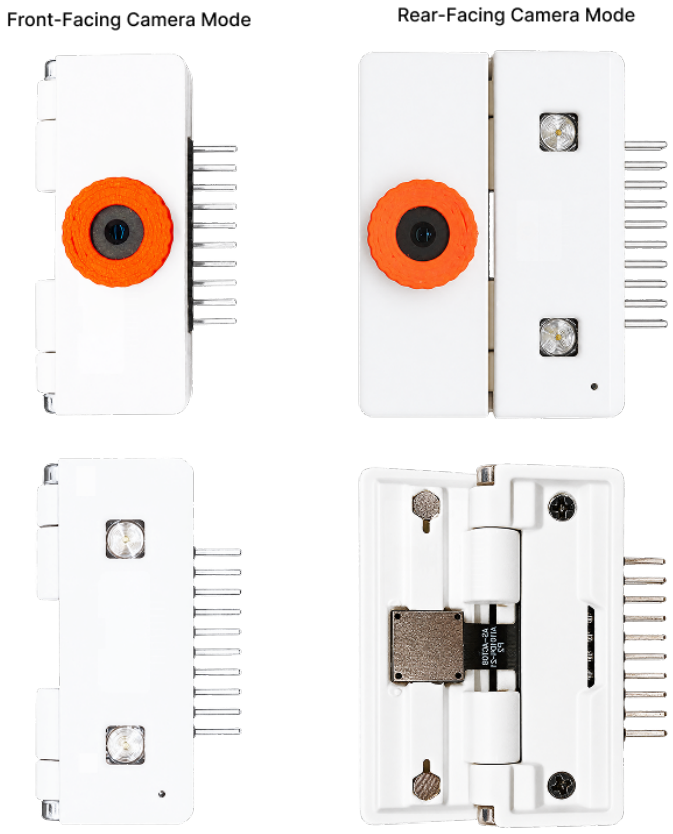

Fig. 37: Camera Module Overview: Front-facing Mode (Folded) on the Left and Rear-facing Mode (Unfolded) on the Right (Click to enlarge)

Required Hardware

- ESP-Mosaico (CoreBoard V1.0 or V1.2)
- ESP-Mosaico Camera Module
- USB cable (data-capable)
- Computer (Windows, Linux, or macOS)

Installation

1. Make sure the host is powered off or the external 3.3 V / 5 V outputs are disabled.
2. Insert the camera module into the host's **left** slot (`H2`), with the orange lens facing up and the flash LEDs facing down. For details, see the ESP-Mosaico User Guide.
3. Power ESP-Mosaico from Type-C and turn it on. The expansion 3.3 V / 5 V rails are output only when GPIO60 is driven low.

The image below shows the camera module connected to the left slot (`H2`) of ESP-Mosaico. Align the module header with the slot and ensure a secure connection.

Fig. 38: Camera Module Connected to ESP-Mosaico (Click to enlarge)

Software Examples The following examples are available for the camera module. Refer to each example's documentation for usage instructions:

- camera_lcd_preview: LCD preview example
- camera_photo_app: Photo capture and gallery example

You can also use the mosaico_module_camera component to develop your own applications.

Hardware Reference

Connector Location The camera module connects to the BaseBoard left slot `H2` through a `2 × 10P` header with a 2.54 mm pitch. The host connector definition is in the Module Interface section of *ESP-Mosaico V1.2*.

DVP and Control Pins The following table applies to both V1.2 and V1.4, which share all pin assignments. `XCLK` is provided by the module oscillator, not by a GPIO on the left slot. See *V1.2 Version* or *V1.4 Version* for the oscillator frequency.

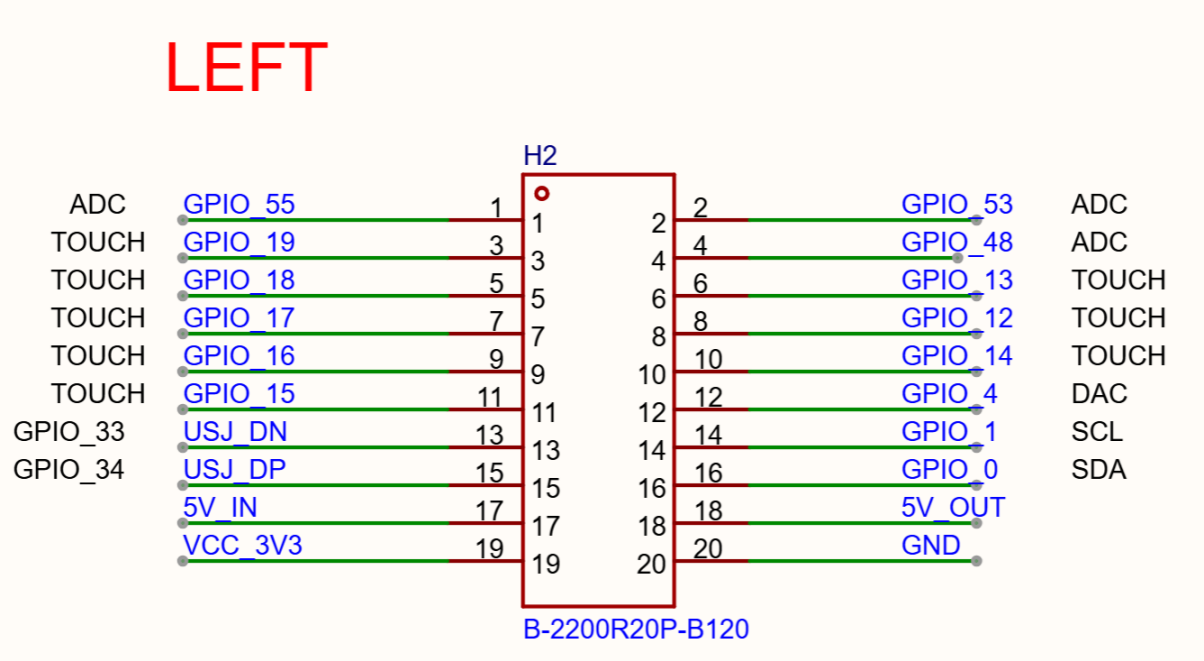

Fig. 39: Left Module Interface Schematic (Click to enlarge)

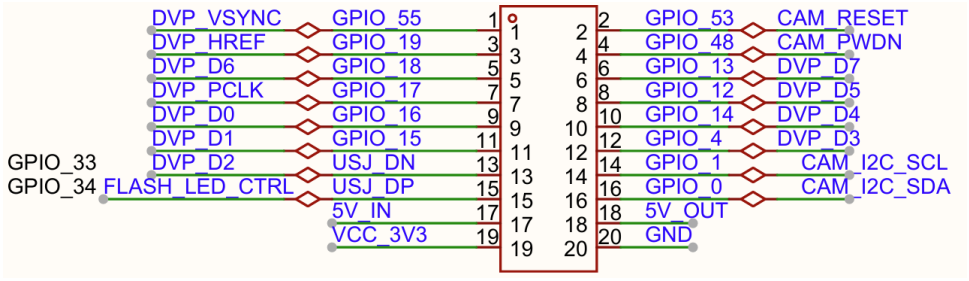

Fig. 40: Camera Module Interface Pinout (Click to enlarge)

Signal	GPIO	Left Slot Pin	Description
DVP_D0	GPIO16	9	Data bit 0
DVP_D1	GPIO15	11	Data bit 1
DVP_D2	GPIO33	13	Data bit 2; multiplexed with USB Serial/JTAG D-
DVP_D3	GPIO4	12	Data bit 3
DVP_D4	GPIO14	10	Data bit 4; reuses the left EEPROM address-select pin
DVP_D5	GPIO12	8	Data bit 5
DVP_D6	GPIO18	5	Data bit 6
DVP_D7	GPIO13	6	Data bit 7
VSYNC	GPIO55	1	Frame sync
DE	GPIO19	3	Data enable
PCLK	GPIO17	7	Pixel clock
RESET	GPIO53	2	Sensor reset
PWDN	GPIO48	4	Sensor power-down
FLASH	GPIO34	15	Flash LEDs; low turns them on, high turns them off; configure as an open-drain output; multiplexed with USB Serial/JTAG D+, off by default
SCCB_SDA	GPIO0	16	Sensor / EEPROM data
SCCB_SCL	GPIO1	14	Sensor / EEPROM clock
VCC_3V3		19	3.3 V supply (controlled by GPIO60)
5V_OUT / 5V_IN		18 / 17	5 V output / input
GND		20	Ground

I2C and Discovery

- Module EEPROM: see each version's description for the model and package. Left-slot address **0x50**, board type **0x07** (`MOSAICO_BOARD_TYPE_CAMERA`).
- Sensor SCCB and the EEPROM share the expansion I2C bus: SDA = GPIO0, SCL = GPIO1. On CoreBoard **V1.2** this is external I2C1; on **V1.0** it is the shared onboard I2C bus.
- After the camera is opened, GPIO14 becomes DVP D4 and EEPROM access to the left slot is suspended until `mosaico_camera_del()` / `bsp_subboard_camera_release()` releases the resource.

Usage Limits

Important:

- **Left slot only**. Inserting the module into the right slot returns `ESP_ERR_NOT_SUPPORTED` and does not configure DVP.
- GPIO33 / GPIO34 are multiplexed with the onboard USB Serial/JTAG PHY. While the camera is claimed, USB Serial/JTAG on the left header is unavailable. The Type-C USB-OTG console is unaffected.
- See the camera specification table for each version's default output format. The software uses four frame buffers by default. Received frames must be returned with `mosaico_camera_return_frame()`, and all frames must be returned before `mosaico_camera_restart()`.
- After power-up, wait for the onboard `XCLK` to stabilize (about 20 ms in the BSP) before accessing SCCB.

Related Documents

- V1.2 Camera Module (OV3640) PCB Schematic (PDF)
- V1.4 Camera Module (SC101IOT) PCB Schematic (PDF)
- OV3640 Datasheet (PDF)
- OV3640 Camera Module Specification (PDF)
- SC101IOT Camera Module Specification (Low-profile Lens) (PDF)
- SC101IOT Camera Module Specification (High-profile Lens) (PDF)

Disclaimer and Copyright Notice

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3.2.4 ESP-Mosaico Multi-function Interaction Module

This guide describes the hardware interface, installation, and software usage of the ESP-Mosaico **Multi-function Interaction Module**. The module connects to the left or right `2 × 10P` module slot and provides ambient-light sensing, PIR motion detection, IR transmit, RGB LEDs, and key / touch input.

Note: The interaction module can be inserted into the left slot (`H2`, EEPROM 0x50) or the right slot (`H1`, EEPROM 0x51). The right slot is mounted 180° relative to the left slot. For the mainboard revision, see *ESP-Mosaico V1.2* or *ESP-Mosaico V1.0*.

The module integrates a GL5506 photoresistor, an AS312 PIR sensor, an IR LED driven by an AO3400A, six WS2812 LEDs (four at the top and two beneath the buttons), and two pads that can be used as mechanical keys or capacitive-touch channels. An AT24C02 EEPROM identifies the module. The board type is 0x16.

This document consists of the following major sections:

- *Getting Started*: Installation, power, and how to use the module.
- *Hardware Reference*: Left/right pin map and peripheral notes.
- *Hardware Revision Details*: Hardware revision history (if any).
- *Related Documents*: Links to related documentation.
- *Disclaimer and Copyright Notice*: Disclaimer and copyright notice.

Getting Started

Description of Components The front view and its main components are shown first, followed by the back view.

Front The following describes the main components on the front by callout number, in clockwise order.

No.	Key Component	Description
1	Touch_R	Right touch pad, usable as a capacitive-touch channel.
2	key_R	Right mechanical key, active-low.
3	LDR	GL5506 photoresistor sampled by ADC for ambient light.
4	PIR Sensor	AS312 PIR sensor that outputs a motion-detect level.
5	IR Emitter	Infrared emitter driven by AO3400A. Default protocol is NEC at 38 kHz.
6	Key_L	Left mechanical key, active-low.
7	Touch_L	Left touch pad, usable as a capacitive-touch channel.
8	WS2812 × 4	Top LEDs LED0–LED3: four addressable RGB LEDs on a single data line.

Back The following describes the main components on the back using the callout numbers in the figure.

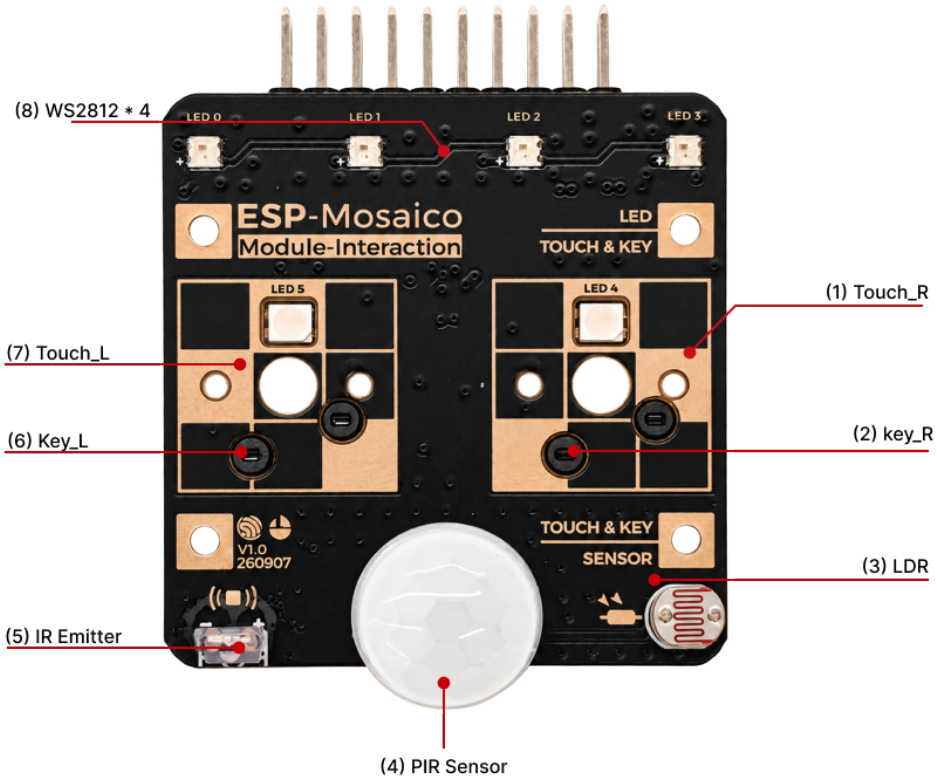

Fig. 41: Interaction module front (Click to enlarge)

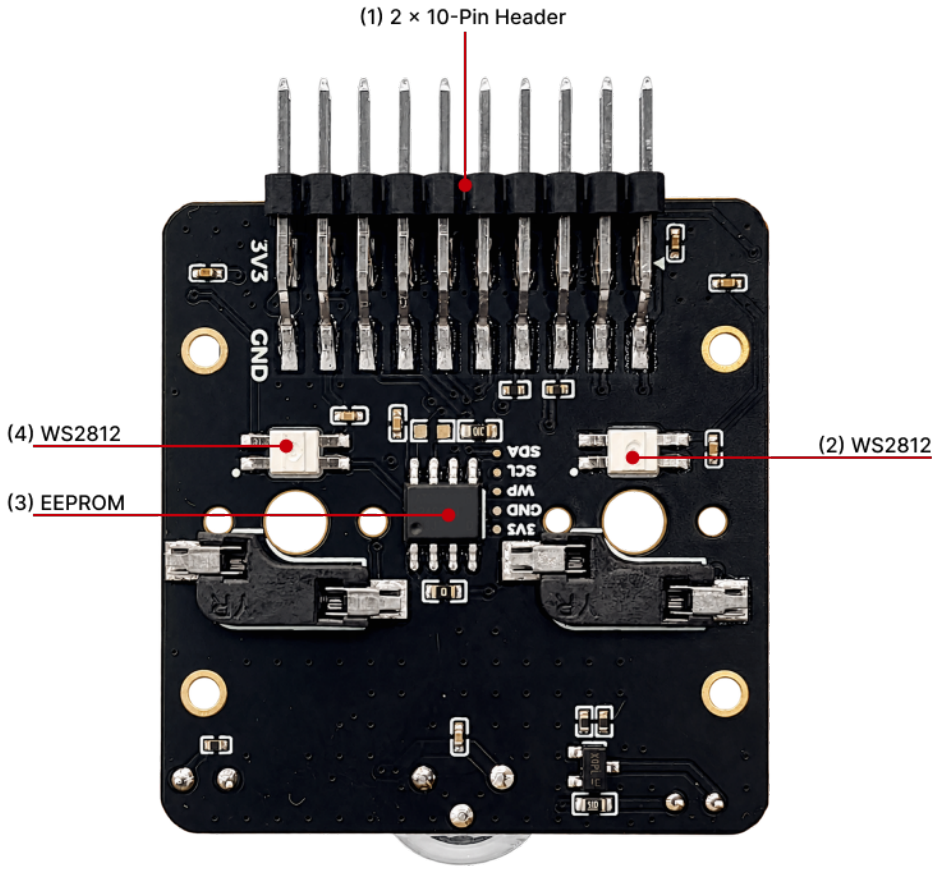

Fig. 42: Interaction module back (Click to enlarge)

No.	Key Component	Description
1	2 × 10-Pin Header	2 × 10P, 2.54 mm pitch header that mates with the BaseBoard left/right module slots.
2	WS2812	Addressable RGB LED on the right side of the figure.
3	EEPROM	AT24C02 module EEPROM. Address 0x50 on the left slot and 0x51 on the right slot. Board type 0x16.
4	WS2812	Addressable RGB LED on the left side of the figure.

Start Using the Module

Required Hardware

- ESP-Mosaico (CoreBoard V1.0 or V1.2)
- ESP-Mosaico Multi-function Interaction Module
- USB cable (data-capable)
- Computer (Windows, Linux, or macOS)

Fig. 43: Interaction module mounting orientation (Click to enlarge)

Installation

1. With the host powered off, or with the external 3.3 V / 5 V outputs disabled, insert the module into the left or right module slot. Rotate the module 180° when using the right slot.

2. Power ESP-Mosaico from Type-C and turn it on. The expansion 3.3 V / 5 V rails are output only when GPIO60 is driven low.
3. After power-on, the host identifies the interaction module from its EEPROM (board type 0x16). A module without this information is not recognized.

How to Use Once the module is recognized, the display shows a control for each inserted slot. Switch among ambient light (LDR), motion (PIR), infrared transmit (IR), RGB LEDs (LED), keys (Key), and touch (Touch). The left and right slots can be used at the same time, and their functions are independent.

Hardware Reference

Slots and Discovery ESP-Mosaico distinguishes the two slots by EEPROM address:

Slot	EEPROM Address	Description
Left (`H2`)	0x50	Address-select GPIO14, driven low
Right (`H1`)	0x51	Address-select GPIO39, driven high; mounted 180° relative to the left slot

Module I2C uses the expansion SDA / SCL pins (GPIO0 / GPIO1). On CoreBoard **V1.2** this is external I2C1; on **V1.0** it is the shared onboard I2C bus.

Left and Right Pin Map The following table lists the interaction-module signals on both slots.

Signal	Left GPIO	Right GPIO	Description
LDR	GPIO53	GPIO46	GL5506 photoresistor ADC
IR	GPIO48	GPIO47	IR transmit drive (AO3400A)
KEY_L / Touch	GPIO13	GPIO11	Left key or touch channel
KEY_R / Touch	GPIO12	GPIO10	Right key or touch channel
PIR	GPIO4	GPIO5	AS312 PIR input
WS2812	GPIO15	GPIO38	Data line for 6 RGB LEDs
SDA	GPIO0	GPIO0	Module I2C / EEPROM
SCL	GPIO1	GPIO1	Module I2C / EEPROM
VCC_3V3	Slot pin 19	Slot pin 19	3.3 V supply (controlled by GPIO60)
5V / GND	Slot pins 17 / 18 / 20	Slot pins 17 / 18 / 20	5 V input/output and ground

The host connector definition is in *ESP-Mosaico V1.2*.

Peripheral Notes

Photoresistor (LDR) The GL5506 is sampled by ADC. The reading is shown as a 0–100 light percentage, which can indicate ambient brightness or drive LED intensity.

PIR Motion Sensor The AS312 outputs a motion-detect level. When motion is detected, the corresponding status on the display lights up.

Infrared Transmit (IR) The IR LED is driven by an AO3400A and sends NEC frames (8-bit address and command) at 38 kHz.

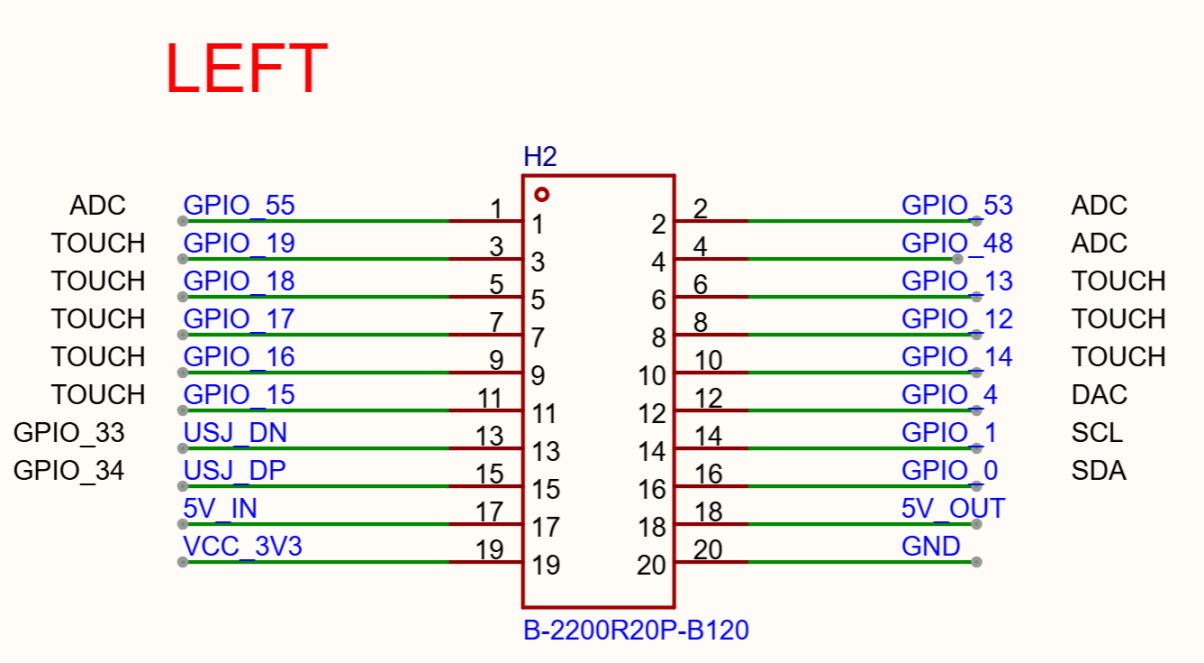

Fig. 44: Left Module Interface Schematic (Click to enlarge)

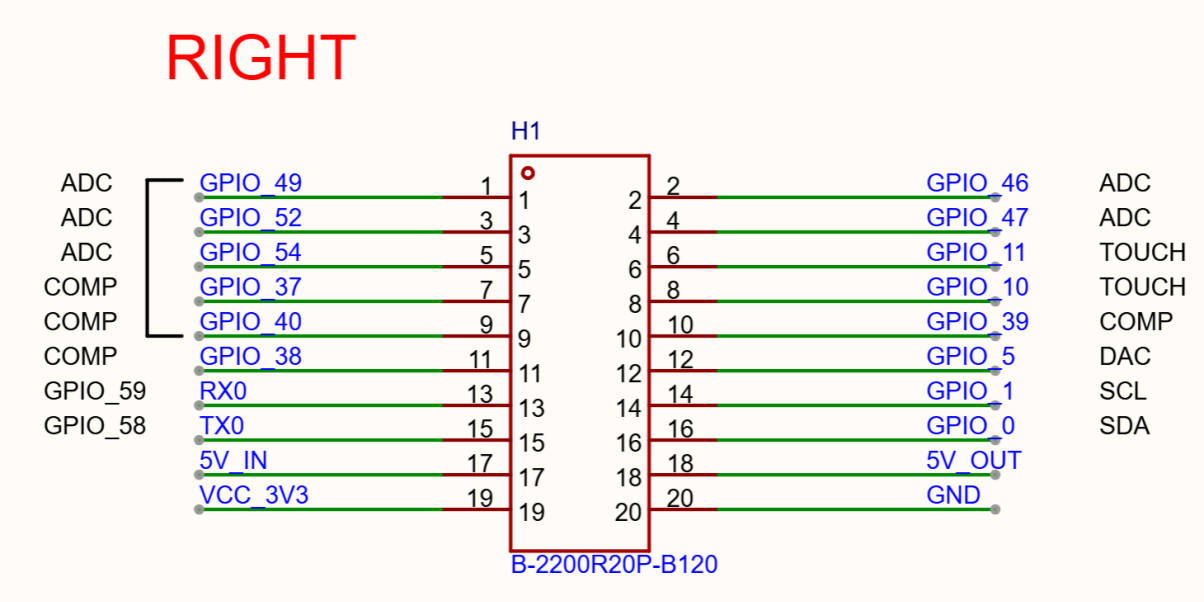

Fig. 45: Right Module Interface Schematic (Click to enlarge)

WS2812 The module has **6** WS2812 LEDs. Colors can be set individually or together, and the brightness can be limited.

Keys and Touch `KEY_L` / `KEY_R` share the same pads and can be used as keys or touch:

- **Key**: active when pressed to ground.
- **Touch**: a finger on the pad triggers it, and pressing the key does as well.

Power The module is powered from the host `VCC_3V3` rail, which is output only when GPIO60 is driven low. On CoreBoard V1.2 the external 3.3 V rail is generated by a dedicated DCDC and is isolated from the internal MCU supply.

Hardware Revision Details

No previous versions.

Related Documents

- ESP-Mosaico Interaction Module Schematic (PDF)

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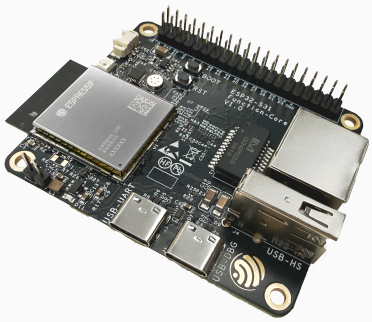
ESP32-S31 Development Boards



ESP32-S31-Function-CoreBoard-1

ESP32-S31-Korvo-1

ESP-Mosaico

Chapter 4

Related Documentation and Resources

4.1 Related Documentation

- ESP32-S31 Datasheet –Specifications of the ESP32-S31 hardware.
- ESP32-S31 Technical Reference Manual –Detailed information on how to use the ESP32-S31 memory and peripherals.
- ESP32-S31 Hardware Design Guidelines –Guidelines on how to integrate the ESP32-S31 into your hardware product.
- ESP32-S31 Product/Process Change Notifications (PCN)
  https://espressif.com/en/support/documents/pcns?keys=ESP32-S31
- ESP32-S31 Advisories –Information on security, bugs, compatibility, component reliability.
  https://espressif.com/en/support/documents/advisories?keys=ESP32-S31
- Certificates
  https://espressif.com/en/support/documents/certificates
- Documentation Updates and Update Notification Subscription
  https://espressif.com/en/support/download/documents

4.2 Developer Zone

- ESP-IDF Programming Guide for ESP32-S31 –Extensive documentation for the ESP-IDF development framework.
- ESP-IoT-Solution Programming Guide - Extensive documentation for the ESP-IoT-Solution development framework.
- ESP-FAQ - A summary document of frequently asked questions released by Espressif.
- ESP-IDF and other development frameworks on GitHub.
  https://github.com/espressif
- ESP32 BBS Forum –Engineer-to-Engineer (E2E) Community for Espressif products where you can post questions, share knowledge, explore ideas, and help solve problems with fellow engineers.
  https://esp32.com/
- The ESP Journal –Best Practices, Articles, and Notes from Espressif folks.
  https://blog.espressif.com/
- See the tabs SDKs and Demos, Apps, Tools, AT Firmware.
  https://espressif.com/en/support/download/sdks-demos

4.3 Products

- ESP32-S31 Series SoCs –Browse through all ESP32-S31 SoCs.
  https://espressif.com/en/products/socs?id=ESP32-S31
- ESP32-S31 Series Modules –Browse through all ESP32-S31-based modules.
  https://espressif.com/en/products/modules?id=ESP32-S31
- ESP32-S31 Series DevKits –Browse through all ESP32-S31-based devkits.
  https://espressif.com/en/products/devkits?id=ESP32-S31
- ESP Product Selector –Find an Espressif hardware product suitable for your needs by comparing or applying filters.
  https://products.espressif.com/#/product-selector

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- See the tabs Sales Questions, Technical Enquiries, Circuit Schematic & PCB Design Review, Get Samples (Online stores), Become Our Supplier, Comments & Suggestions.
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Chapter 5

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